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RESEARCH ARTICLE

# Smart Mobile Features for Hajj: From Pilgrim Challenges to Feasible Solutions

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**ABSTRACT** The Hajj pilgrimage is one of the largest and most significant religious gatherings in the world, attracting millions of Muslims belonging to various backgrounds, annually to Makkah and Madinah. Despite its spiritual significance, the event poses a range of challenges such as logistical, navigational, communication, and safety etc. Addressing these challenges requires a mobile application consisting of state-of-the-art technological solutions. This study proposes a Delphi-guided methodology to identify, evaluate, and prioritize mobile application features aimed at supporting Hajj pilgrims. The process began by identifying core challenges faced by pilgrims and proposing 15 features along with its mapping with the identified challenges. These features were then evaluated by a panel of nine domain experts across two phases. The first phase assessed the features' relevance to addressing pilgrim challenges, while the second phase evaluated each feature across four feasibility dimensions: infrastructure, development, regulatory compliance, and environmental suitability. The study presents a validated and prioritized set of mobile application features, offering actionable guidance for developers, policymakers, and Hajj organizers. These findings support the design of a mobile application that can enhance the Hajj experience by improving communication, navigation, safety, and accessibility.

**INDEX TERMS** Hajj, mobile application, smart features, pilgrim challenges, feature prioritization, Delphi method, feasibility analysis.

## I. INTRODUCTION

The Hajj and Umrah pilgrimages are among the most significant religious and cultural events globally, attracting millions of Muslims annually to the holy cities of Makkah and Madinah [1], [2]. These large gatherings create a range challenges related to logistics, navigation, communication, and safety, especially because pilgrims come from a wide range of languages, cultures, and backgrounds [3], [4]. Ensuring a smooth and safe pilgrimage experience requires a validated set of state-of-the-art technology-supported features in the mobile application that can provide real-time multilingual communication, spatial navigation, emergency coordination, and crowd management.

Despite recent advances in digital technologies and growing interest in smart pilgrimage management, existing research largely focuses on isolated technological components or standalone applications. Such fragmented approaches often fail to meet the holistic and evolving needs of pilgrims throughout the Hajj journey [2], [4]. Consequently, a clear research gap exists in developing integrated, user-centered frameworks that systematically identify, evaluate, and justify mobile solutions tailored to the unique conditions and constraints of the pilgrimage context.

Emerging technologies—such as Artificial Intelligence (AI), Augmented Reality (AR), Geographic Information Systems (GIS), Natural Language Processing (NLP), and Optical Character Recognition (OCR)—offer powerful ways to improve digital support for large-scale gatherings [5], [6]. For example, AR can help pilgrims find their way and understand their surroundings in crowded or unfamiliar areas [7], [8]. Recent studies also explore AR's role in religious heritage management using immersive experiences for visitor engagement [9], and in promoting sustainability

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at cultural heritage attractions [10]. NLP-based translation tools can overcome language barriers among pilgrims from different countries [11], [12], while GIS applications can support real-time crowd management and help authorities make quick spatial decisions [13], [14]. Recent tourism studies further highlight travelers' growing reliance on machine translation services during international trips [15].

However, the effective integration of these technologies into a single mobile application for Hajj support remains underexplored. This study aims to fill this gap by proposing a Delphi-guided methodology to identify, evaluate and prioritize the features of mobile applications designed to support Hajj pilgrims. The novel contributions of this research are as follows:

- It identifies core challenges faced by pilgrims during the Hajj.
- It proposes a set of 15 mobile application features directly mapped to these challenges.
- It presents a Delphi-guided methodology for identifying and assessing mobile features through input from domain experts across multiple dimensions.
- It offers a prioritized list of features, providing clear guidance for developers, policymakers, and technology planners.

The remainder of the paper proceeds as follows: Section II provides an overview of related literature and existing solutions. Section III describes the methodology. Section IV describes the proposed smart features in detail. Section V mentions the results. Section VI provides an in-depth discussion, covering result interpretation, practical implications, limitations, and directions for future research. Finally, Section VII concludes the paper by summarizing key contributions and findings.

## II. RELATED WORK

This section reviews prior research across multiple domains relevant to enhancing the Hajj pilgrimage experience. The literature is organized into three key thematic areas: (1) Simulation and Decision Support Technologies for Hajj Management, (2) Mobile Applications and Smart Solutions for Pilgrim Support, and (3) Surveys and Empirical Insights within the Hajj Technology Domain.

### A. SIMULATION AND DECISION SUPPORT TECHNOLOGIES FOR HAJJ MANAGEMENT

Several researchers have primarily focused on infrastructure-level simulations and decision-support systems to manage crowd flow during Hajj rituals. While these approaches contribute significantly to macro-level planning, they often lack user-centric adaptability and real-world deployment. For instance, Haase et al. [16] developed an Operations Research-based Decision Support System (ORDSS) that integrates scheduling algorithms with real-time video analytics to enhance safety during high-density rituals such as Rami-al-Jamarat. While effective in managing infrastructure-level decisions, the solution remains resource-intensive and lacks adaptability to individual user contexts. Similarly, Yaagoubi et al. [13] employed Geographic Information Systems (GIS) and agent-based modeling to simulate pilgrim movements and optimize scheduling flows. Despite offering robust simulation capabilities, the system does not account for real-time environmental variability or the personalized needs of pilgrims. Wulansari [17] explored crowd dynamics using steering behavior algorithms in augmented environments, specifically during the Tawaf ritual. Though the approach enables realistic simulation and animation, its primary focus remains academic, lacking real-world deployment or attention to user experience. In contrast, our study adopts a pilgrim-centered approach by identifying and prioritizing mobile application features based on expert evaluations. This ensures both contextual feasibility and practical applicability, aiming to directly enhance the on-ground experience of Hajj pilgrims.

### B. MOBILE APPLICATIONS AND SMART SOLUTIONS FOR PILGRIM SUPPORT

Ilias et al. [18] developed an Android application featuring GPS tracking and a Tawaf counter. While technically functional, the app followed the ADDIE instructional design model but lacked iterative user feedback and did not consider regulatory constraints, thereby limiting its scalability and contextual relevance. Yamin et al. [19] proposed leveraging integrated social media platforms to enhance communication and emergency coordination among pilgrims. However, their concept remained at the proposal stage, lacking implementation and user-centric validation. In another direction, Taileb et al. [8] and Albaqami et al. [7] introduced augmented reality (AR)-based solutions to aid spatial navigation and crowd guidance in Makkah. Though promising, these applications remain at the conceptual or prototype stage with limited real-world testing. Al-Shaery et al. [20] proposed a novel AI system that uses wearable physiological sensors to monitor pilgrim fatigue, emotional state, and activity levels. While innovative, the system was tested on a small sample and lacks evidence of generalizability or user acceptance. In contrast, our study presents a Delphi-guided approach that evaluates and proposes mobile features that offer a more comprehensive solution.

### C. SURVEYS AND EMPIRICAL INSIGHTS WITHIN THE HAJJ TECHNOLOGY DOMAIN

Felemban et al. [1] presented a comprehensive survey and proposed a taxonomy covering technologies such as computer vision, wireless networks, and spatial computing for crowd management during Hajj. While their work effectively outlines the technical landscape, it lacks a structured mechanism for validating specific features from a user-centric perspective. Showail [3] conducted an extensive review of over 33 years of research on the application of information and communication technologies (ICT) to address various challenges in Hajj and Umrah. The study categorizes

efforts into ten application areas and identifies over thirty distinct technologies, organizing research by both use case and technology. Shah [2] explored the role of artificial intelligence in crowd management, ritual optimization, and infrastructure support. Although the classification is wide-ranging, it lacks detailed assessment of individual mobile features and their feasibility for real-world deployment. Shambour and Gutub [4] and Binsawad and Albahar [21] offered analytical reviews of IoT and ICT applications in Hajj and Umrah, focusing on infrastructure and service delivery. Similarly, Khan and Shambour [22] evaluated mobile applications available on Google Play and found that most rely on basic services like Hajj guides and video streaming, with limited interactivity and feature diversity. In contrast, our work goes beyond broad surveys by proposing a Delphi-guided methodology for identifying, validating, and prioritizing mobile application features. This structured approach integrates expert consensus to ensure that the proposed features effectively address pilgrim challenges while supporting validated design and implementation feasibility.

## III. METHODOLOGY

Our research methodology is guided by the Delphi technique [23], [24], [25], [26], [27], [28], a structured method designed to systematically gather expert opinions and build consensus on complex and multifaceted issues. The primary objective of applying the Delphi-based approach in this study is to reach a reliable consensus among domain experts regarding the essential features of a mobile application intended to support Hajj pilgrims. Figure 1 summarizes our Delphi-guided methodology in a concise, stepwise flow. Step 1 identifies pilgrims' challenges (CH1–CH9). Step 2 derives the candidate features, benchmarks them against the Nusuk app, specifies real-world use cases and implementation details, and maps each feature to the relevant challenges for clarity. Step 3 develops concise conceptual illustrations to aid expert judgment of how each feature would look and function. Step 4 establishes challenge-relevance consensus via Delphi rounds. Step 5 evaluates feasibility (infrastructure, development, regulatory, environmental) and yields a prioritized feature set. Importantly, each step can use the accumulated outputs from all prior steps. We describe each step in detail in the subsequent sections.

### A. STEP 1: IDENTIFICATION OF PILGRIM CHALLENGES

To identify the key challenges faced by Hajj pilgrims, we began with a thorough literature review. Notable sources include studies and evaluations that highlight recurring issues encountered during previous Hajj seasons [3], [29], [30], [31], [32].

In addition to reviewing the literature, the authors also drew upon their own professional experience with Hajj-related applications and engaged with domain experts actively involved in projects for the Ministry of Hajj and Umrah. These experts not only possess hands-on experience with digital technologies used during the pilgrimage but have also performed Hajj themselves, giving them a unique practical perspective. This approach—combining documented research with experiential and expert insights—enabled the authors to compile a comprehensive list of the most prevalent challenges, as summarized in Table 1.

**TABLE 1. Common challenges faced by pilgrims during hajj.**

| Code | Challenge |
|---|---|
| CH1 | Language barriers and communication issues |
| CH2 | Navigational difficulties |
| CH3 | Inaccessibility of reliable and timely information |
| CH4 | Disorientation or separation from group members |
| CH5 | Difficulties during emergency situations (e.g., stampede, heat-stroke, fire) |
| CH6 | Lack of awareness about religious rituals and their significance |
| CH7 | Inability to access digital services due to internet unavailability |
| CH8 | Confusion related to transport options, stops, and schedules |
| CH9 | Accessibility issues for illiterate or visually impaired pilgrims |

### B. STEP 2: FEATURE IDENTIFICATION

Building upon the challenges identified in Step III-A, a preliminary list of mobile application features was proposed to address the specific needs of Hajj pilgrims. This feature set was developed through a structured three-step process:

First, we conducted an in-depth literature review, performed a comparative analysis of existing Hajj applications, and incorporated insights from the authors' prior experience in Hajj-related technologies. Relevant details of the literature review are provided in Section II and Section III-A. Among the notable applications, *Nusuk* is the official and currently only active application of the Ministry of Hajj and Umrah, with more than "10+M" downloads on Google Play and frequent updates. Accordingly, we benchmarked our proposed features primarily against *Nusuk* (see Table 2) to clarify which needs are already addressed and where gaps remain—insights that directly motivated our Delphi-guided feature selection. Other apps—such as Manasikana,[1] Eatmarna, and Hajj Navigator[2]—appear to be legacy or low-activity with limited recent documentation. Based on this analysis, an initial set of features was identified and refined through internal discussions.

Second, we conducted a detailed examination of each proposed feature to identify real-world use cases and potential implementation pathways. This step aimed to understand how each feature might function in practice, including where and when it would be most beneficial to pilgrims. The implementation analysis involved identifying suitable tools, frameworks, APIs, and device capabilities necessary for effective deployment. These details are outlined in Section IV. Providing this information allowed evaluators to better assess the features from technical and developmental feasibility perspectives.

Third, after establishing the preliminary feature list, we carried out a feature-to-challenge mapping exercise. The

[1] https://manasikana.softonic-ar.com/android
[2] https://play.google.com/store/apps/details?id=com.phvg.hajjnavigator

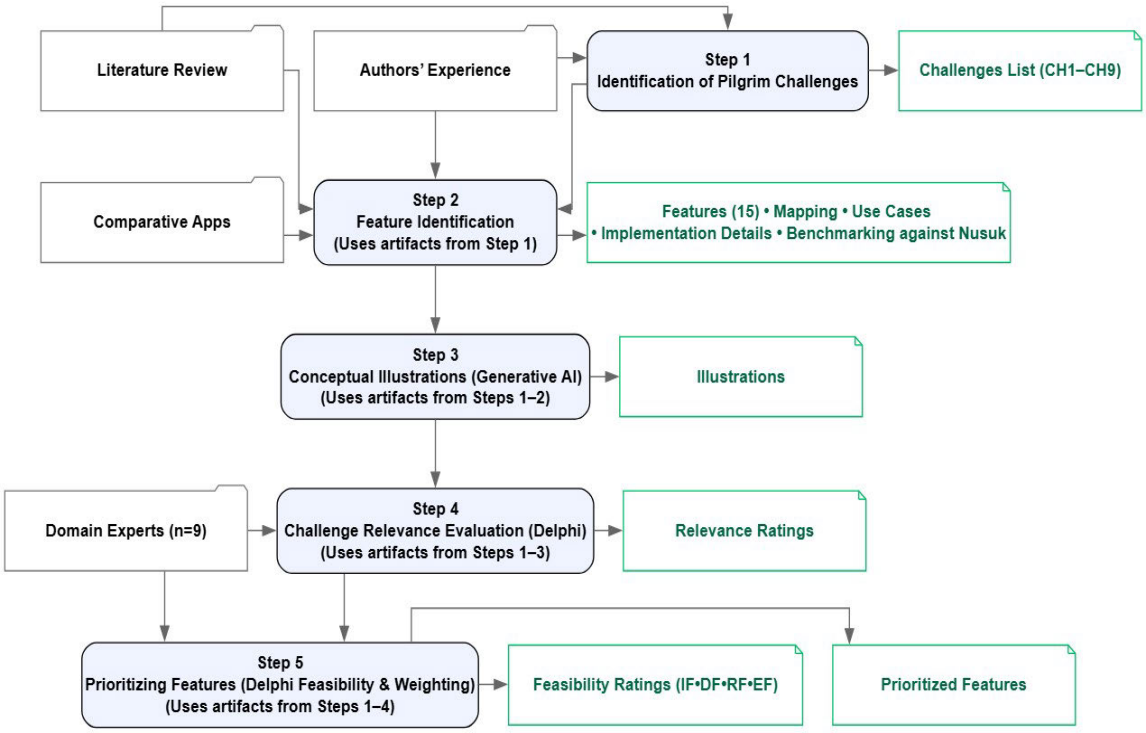



**FIGURE 1.** Overview of the delphi-guided methodology.

**TABLE 2.** Proposed mobile app features—benchmarking against the nusuk app.

| Features | Nusuk |
|---|---|
| **C1. Multilingual Translation Services** | |
| Text-to-Text Translation | — |
| Voice-to-Text Translation | — |
| Text-to-Voice Translation | — |
| Text or Voice to Image or Map Translation | — |
| **C2. Offline and Integrated AI-Based Services** | |
| Offline Guide Maps | ◐ |
| Multilingual AI ChatBot | — |
| **C3. Augmented Reality (AR)-Based Solutions** | |
| AR PathFinder | — |
| Lost Pilgrim Tracker | — |
| MeetPoint Finder | — |
| Emergency Guide | ◐ |
| Virtual Guide to Holy Sites | ◐ |
| Scan and Understand | — |
| **C4. Additional Support Services** | |
| Charity Information and Locations | ◐ |
| Transportation and Navigation | ✓ |
| Haramain Explorer | ◐ |

*Legend:* ✓ present; ◐ similar/partial; — not stated

goal of this step was to evaluate how well each feature addresses the real-world challenges faced by Hajj pilgrims and to determine the coverage and effectiveness of the overall feature set. By explicitly aligning each feature with one or more identified challenges, we could assess its potential impact—specifically, the number and types of challenges it could mitigate. A summary of the feature-to-challenge alignment is presented in Table 3, where a checkmark (✓) indicates that a feature directly addresses a particular challenge.

Following this three-step process, we finalized a total of 15 features. These features are summarized in Table 2. For clarity and organization, the features have been categorized into four thematic groups based on their technological focus such as (i) C1. Multilingual Translation Services, (ii) C2. Offline and Integrated AI-Based Services, (iii) C3. Augmented Reality (AR)-Based Solutions, (iv) C4. Additional Support Services. Comprehensive descriptions of each feature—including use cases, and implementation considerations provided in Section IV.

### C. STEP 3: CONCEPTUAL ILLUSTRATIONS USING GENERATIVE AI

To support expert understanding and strengthen feature evaluation, we adopt a practice increasingly used in HCI/UX research—employing AI-generated conceptual illustrations as communication aids to visualize concepts and ideas [33], [34], [35]. These visuals translate high-level ideas into concrete, high-fidelity screens within minutes, reduce ambiguity in user flows, and enable rapid iteration before costly prototyping. We applied this approach to create visual representations of how each proposed feature might appear in real Hajj scenarios. Several illustrations were produced with generative-AI tools (e.g., ChatGPT via image prompts); when a suitable image could not be generated, we sourced publicly available images and manually edited them to reflect the intended concept. For example, the illustration of the *Scan and Understand* feature was customized using such an approach to depict how Arabic text in the environment would be scanned and translated for the user. Along with those conceptual illustrations, the textual descriptions, use case examples, implementation details, all of them help in understanding the functionality and purpose of these features.

### D. STEP 4: EVALUATING FEATURES BASED ON ADDRESSING CHALLENGES

To determine whether all 15 proposed features should be included in the mobile application, a Delphi-based evaluation was conducted with a panel of nine domain experts. Although there is some debate regarding the appropriate number of experts participated in Delphi studies, the standard range commonly reported in the literature is between 8 and 23 participants [36]. Moreover, there are studies where fewer than nine participants were involved, such as [37], which employed only four participants, and [38], which used eight participants. Therefore, the inclusion of nine experts in our Delphi study aligns with established practices and falls within the acceptable range supported by the literature. The primary goal of this evaluation was to assess how effectively each feature addresses the challenges outlined in Table 3. The process consisted of two rounds, enabling iterative reflection and fostering consensus among participants.

All nine experts selected for this phase had prior experience performing Hajj and possessed relevant technological expertise through their professional work with organizations in Saudi Arabia. In the first round, experts were provided with a comprehensive overview of the proposed features, including the mapping between features and challenges (as shown in Table 3), conceptual illustrations generated using generative AI tools, and key implementation considerations. Each expert was then asked to rate the extent to which each feature addressed pilgrims' challenges using a 5-point likert scale (1 – Merely addresses pilgrims' challenges, 5 – Fully addresses pilgrims' challenges).

In the second round, experts whose ratings significantly diverged from the group's average were shown the aggregated responses of their peers. They were invited to revise their ratings, if necessary, after reconsideration. This iterative feedback loop helped refine the evaluations and strengthened the overall consensus. The insights gathered from this process formed the basis for making informed decisions about which features to include in the final application design.

### E. STEP 5: PRIORITIZING FEATURES

To determine which of the 15 proposed features should be prioritized for the development, nine domain experts were engaged to evaluate each feature based on four key dimensions of feasibility. These dimensions were carefully selected to capture both the technical practicality and the contextual relevance of each feature within the specific setting of the Hajj pilgrimage.

Infrastructure feasibility referred to how realistically a feature could be deployed using on-premise infrastructure, to ensure data security and full administrative control. Development feasibility assessed the relative ease and speed with which the feature could be implemented, taking into account existing tools and capabilities. Regulatory feasibility examined the degree to which the feature complied with Saudi Arabia's legal frameworks and official Hajj protocols. Environmental feasibility considered the functionality and usability of the feature under actual Hajj conditions, which are often characterized by high crowd density, slow internet speed, and extreme climate.

Each expert rated the features on a five-point likert scale for every dimension. A score of 1 indicated significant barriers—whether infrastructure, developmental, regulatory, or environmental—while a score of 5 suggested minimal challenges and high compatibility. For example, in the context of infrastructure feasibility, a score of 1 indicated that a feature would be extremely difficult to deploy using on-premise infrastructure, whereas a score of 5 signified that deployment would be straightforward. A score of 1 in development feasibility reflected a highly time-consuming process, whereas a 5 implied rapid implementation. Similarly, a score of 1 in regulatory feasibility pointed to major legal or compliance issues, while a 5 signified alignment with existing policies. In environmental feasibility, a score of 1 denoted severe limitations under real Hajj conditions, while a 5 suggested the feature would function effectively in those circumstances.

The evaluation was conducted in two iterative rounds to allow experts to reflect on their assessments and promote convergence toward consensus. After the second round, average scores across all experts were calculated for each dimension per feature. To synthesize these evaluations into a single prioritization score, weighted averages were applied, allowing greater emphasis on dimensions deemed strategically important for the smart hajj application. These final scores provide a data-driven foundation to help guide stakeholders and development teams in focusing their efforts on features that are both impactful and realistically implementable within the operational context of Hajj.

**TABLE 3.** Mapping of features to pilgrims' challenges.

| Feature | CH1 | CH2 | CH3 | CH4 | CH5 | CH6 | CH7 | CH8 | CH9 | CH10 |
|---|---|---|---|---|---|---|---|---|---|---|
| Text-to-Text Translation | ✓ | | | | | ✓ | | | | ✓ |
| Voice-to-Text Translation | ✓ | | | | | | | | | ✓ |
| Text-to-Voice Translation | ✓ | | | | | ✓ | | | | ✓ |
| Text/Voice to Image/Map Translation | ✓ | ✓ | | | | | | | | ✓ |
| Offline Guide Maps | | ✓ | ✓ | | | | ✓ | | | |
| Multilingual AI Chat-Bot | ✓ | | ✓ | | ✓ | ✓ | ✓ | | | |
| AR PathFinder | | ✓ | | | | | | | ✓ | |
| Lost Pilgrim Tracker | | | | ✓ | | | | | | |
| MeetPoint Finder | | | | ✓ | | | | | | |
| Emergency Guide | | ✓ | ✓ | | ✓ | | ✓ | | | |
| Virtual Guide to Holy Sites | | | ✓ | | | ✓ | | | | |
| Scan and Understand | ✓ | | ✓ | | | ✓ | | | | ✓ |
| Charity Info and Locations | | | ✓ | | | | | ✓ | | |
| Transportation and Navigation | | ✓ | ✓ | | | | ✓ | | ✓ | |
| Haramain Explorer | | ✓ | ✓ | | | ✓ | | | | |

## IV. PROPOSED SMART FEATURES

Building upon a comprehensive literature review, analysis of existing Hajj-related mobile applications, and insights from domain experts—as outlined in Section III-B—this section presents a set of 15 proposed mobile application features aimed at addressing the key challenges faced by Hajj pilgrims (refer to Section III-B). Each feature is accompanied by representative use cases (summarized in Table 4) and implementation considerations, including recommended tools, libraries, and APIs (detailed in Table 5). These features have been categorized into four groups based on their focus on cutting-edge technologies such as (i) Multilingual Translation Services, (ii) Offline and AI-Based Services, (iii) Augmented Reality (AR)-Based Solutions and (iv) Additional Support Services. The details of each feature are presented in the following sections.

### A. MULTILINGUAL TRANSLATION SERVICES

Language differences often create major communication challenges during international pilgrimages. To address this issue, the proposed solution will include advanced multilingual translation features within a single, easy-to-use mobile application. These translation services will help pilgrims interact more effectively, understand important information, and navigate their surroundings regardless of their native language.

#### 1) TEXT-TO-TEXT TRANSLATION

Text-to-text translation allows pilgrims to convert written content from one language to another with high contextual accuracy. This feature is powered by transformer-based Neural Machine Translation (NMT) models such as Google's T5 (Text-to-Text Transfer Transformer) [39] and Facebook's M2M-100 [40], which support direct translation between multiple language pairs without relying on English as an intermediary. These models can be fine-tuned using domain-specific data to improve relevance and translation quality within the Hajj context. The mobile application interface enables users to input or paste text and receive translated output in their preferred language. This helps pilgrims comprehend signage, medical forms, safety announcements, and religious instructions. Figure 2 illustrates a sample interface.




**FIGURE 2.** Text-to-text translation app interface for pilgrims.

##### a: USE CASE

For example, a non-Arabic-speaking pilgrim navigating Mina may encounter signs or medical forms only available

**TABLE 4.** Summarized use cases for proposed features.

| Feature | Representative Use Case |
|---|---|
| Text-to-Text Translation | Pilgrim receives a written message or sign in a foreign language and wants to understand it in their native language. |
| Voice-to-Text Translation | A pilgrim speaks a question in their native language and receives a written translated message in Arabic or English. |
| Text-to-Voice Translation | A visually impaired or elderly pilgrim uses the app to listen to text instructions in their preferred language. |
| Text/Voice to Image/Map Translation | A user says "Where is the nearest restroom?" and receives a visual map indicating its location. |
| Offline Guide Maps | A pilgrim loses signal but navigates safely using pre-downloaded offline maps to return to their camp in Mina. |
| Multilingual AI ChatBot | A pilgrim asks, "How do I perform Rami?" and receives step-by-step ritual instructions in Urdu. |
| AR PathFinder | A pilgrim holds up their phone and sees AR overlays pointing toward less crowded paths during Tawaf. |
| Lost Pilgrim Tracker | A group member uses AR and BLE signals to visually track and locate a separated pilgrim in a crowd. |
| MeetPoint Finder | A pilgrim follows AR arrows on-screen to navigate back to their group's predefined meeting point. |
| Emergency Guide | In a medical emergency, the app guides a user in real-time via AR to the nearest first aid station. |
| Virtual Guide to Holy Sites | Before arriving at Hajj, a user explores 3D models of key ritual sites to understand movement and layout. |
| Scan and Understand | A pilgrim scans Arabic signage at a bus terminal and instantly sees the English translation overlaid in AR. |
| Charity Information and Locations | A pilgrim filters for "food" services and is navigated to the nearest charity offering hot meals. |
| Transportation and Navigation | During Tafweej, pilgrims are alerted about optimal bus timings and get guided to pickup points. |
| Haramain Explorer | A pilgrim looks up the Cave of Hira, reads historical details, and navigates there using map integration. |

in Arabic. Using this feature, phrases in Arabic can be quickly translated into Urdu, Bahasa, English, or other languages. Similar functionalities have been implemented in humanitarian settings through apps like Tarjimly, which support cross-language communication for refugees and aid workers.[3]

#### *b: IMPLEMENTATION CONSIDERATIONS*

To optimize performance, models can be run on-device using lightweight frameworks such as LiteRT[4] or ONNX Runtime.[5] For enhanced accuracy and reduced latency, cloud-based solutions like Google Cloud Translation API[6] or Microsoft Azure Translator[7] can be utilized. The interface should support source/target language selection, automatic language detection, spell-checking, and offline translation capabilities.

### 2) VOICE-TO-TEXT TRANSLATION

This feature facilitates real-time verbal communication by transcribing spoken input into translated text. It combines Automatic Speech Recognition (ASR) and NMT to enhance accessibility in crowded or multilingual environments.




**FIGURE 3.** Voice-to-text translation app interface for pilgrims.

#### *a: USE CASE*

A typical use case involves a pilgrim requiring medical assistance but unable to communicate with Arabic-speaking staff. The user can speak into the app in their native language—such as French or Urdu—and have it transcribed and translated into Arabic text. This enables two-way communication by reversing the translation process. Similar

[3]https://play.google.com/store/apps/details?id=com.tarjimlymobile&hl=en
[4]https://ai.google.dev/edge/litert
[5]https://onnxruntime.ai/
[6]https://cloud.google.com/translate
[7]https://azure.microsoft.com/en-us/services/cognitive-services/translator/

capabilities have been successfully deployed in airports and aid settings through apps like Google Translate and Tarjimly.

#### b: IMPLEMENTATION CONSIDERATIONS

For ASR, models such as Google Speech-to-Text[8] and Mozilla DeepSpeech[9] can be employed. These tools can be adapted to handle background noise—prevalent in Hajj environments—through noise-robust training and advanced microphone integration. After transcription, the output is passed to the same NMT pipeline described in the previous subsection. The mobile interface can support single-tap microphone activation, live display of translated text, auto-detection of input language, and history tracking of past interactions.

### 3) TEXT-TO-VOICE TRANSLATION

Text-to-Voice Translation enhances accessibility by converting written text into spoken language using Text-to-Speech (TTS) technology. This feature is particularly valuable for visually impaired individuals, elderly pilgrims, or users with low literacy. By enabling users to listen to important information—such as instructions, navigation directions, and alerts—this function promotes inclusivity and usability in high-mobility, information-dense environments like pilgrimage zones. As shown in Figure 4, users can tap on any text or enter a sentence to hear it read aloud in a natural-sounding voice tailored to their selected language.




**FIGURE 4.** Text-to-voice translation interface for pilgrims.

#### a: USE CASE

Pilgrims who are unable to read on-screen instructions—due to visual impairments, age-related limitations, or language unfamiliarity—may struggle to interpret important updates, such as changes in group schedules or health advisories. With this feature, a pilgrim can listen to such information in their native language (e.g., Urdu, Bengali, or Bahasa) rather than relying on others for assistance. It is equally beneficial in noisy or visually cluttered environments like transport stations or crowded tents. This approach has already been proven in various domains including screen readers, digital assistants like Siri and Alexa [41].

#### b: IMPLEMENTATION CONSIDERATIONS

This feature can leverage neural TTS models such as Tacotron 2 [42] or FastSpeech 2 [43] to produce highly natural and multilingual speech. These can be deployed using cloud-based services like Google Cloud Text-to-Speech[10] or Amazon Polly[11] for high-quality output. Alternatively, offline functionality can be supported through lightweight on-device models using LiteRT[12] or ONNX Runtime.[13] The user interface should offer one-tap playback, voice customization, and adjustable playback speed to support different user preferences and accessibility needs.

### 4) TEXT OR VOICE TO IMAGE/MAP TRANSLATION

This feature combines Natural Language Processing (NLP) and Geographic Information Systems (GIS) to convert user queries—whether typed or spoken—into dynamic, visual representations such as annotated maps or location-specific images. It supports wayfinding by reducing language barriers and enabling intuitive navigation within complex, crowded environments like those encountered during Hajj. As shown in Figure 5, the interface can interpret natural language inputs such as "Where is the nearest restroom?" and present visual directions or highlighted locations on a map.

#### a: USE CASE

Pilgrims often struggle to locate essential services such as medical stations, restrooms, or their group's tent—especially in areas like Mina. For instance, when a user inputs "Where is the nearest clinic?" the system should identify their location and return a nearby clinic's location on a map, along with a route from their current position. This eliminates the need for manually searching through dense maps or relying on verbal instructions. Real-world navigation platforms like Google Maps and Apple Maps already integrate voice-based querying, demonstrating the practical feasibility of this concept [44].

#### b: IMPLEMENTATION CONSIDERATIONS

Voice queries can be processed using ASR systems like Google Speech-to-Text[14] or Mozilla DeepSpeech.[15] The text output is then interpreted using NLP models such as

[8] https://cloud.google.com/speech-to-text
[9] https://github.com/mozilla/DeepSpeech
[10] https://cloud.google.com/text-to-speech
[11] https://aws.amazon.com/polly/
[12] https://ai.google.dev/edge/litert
[13] https://onnxruntime.ai/
[14] https://cloud.google.com/speech-to-text
[15] https://github.com/mozilla/DeepSpeech




**FIGURE 5.** Text/voice-to-map translation interface for pilgrims.

BERT [45] or Google T5 [39], which extract location entities and determine user intent.

GIS platforms such as Google Maps API[16] or OpenStreetMap (OSM)[17] can then be used to render location-based visual content. The system can display annotated maps, images of landmarks, and optimal routes. Offline functionality can be achieved by caching essential map tiles locally. The interface should support interactive features such as tapping on points of interest for additional details and offering both visual and text-based instructions.

### B. OFFLINE AND INTEGRATED AI-BASED SERVICES

Given the intermittent internet connectivity often encountered during Hajj, the proposed system incorporates AI-driven services capable of functioning effectively in offline or low-connectivity scenarios. This ensures uninterrupted access to essential information and guidance, thereby enhancing the autonomy and safety of pilgrims.

#### 1) OFFLINE GUIDE MAPS

Offline guide maps are designed to provide pilgrims with high-resolution, interactive visualizations of key religious sites—including Al-Haram, Mina, Arafat, and Muzdalifah—as well as pedestrian routes, emergency facilities, and accommodation zones. These maps are fully functional without an internet connection and aim to support independent navigation in congested or signal-restricted environments. The map interface can enable users to zoom, pan, and access annotated overlays such as medical tents, ritual stations, and walking paths. Layered views can be customized to suit the user's current location or needs. Figure 6 presents a conceptual illustration that helps enhance understanding of the feature's functionality.



**FIGURE 6.** Offline guide maps interface for pilgrims.

##### a: USE CASE

In locations like Mina or Arafat, pilgrims may experience weak or no cellular service. For instance, if a pilgrim is separated from their group or requires medical assistance, they can rely on offline maps to navigate to the nearest tent or emergency facility. Such functionality has already proven effective in outdoor tourism and humanitarian relief apps like Maps.me and OsmAnd.[18]

##### b: IMPLEMENTATION CONSIDERATIONS

The system can utilize mobile-optimized Geographic Information System (GIS) frameworks such as Mapbox GL Native[19] or Leaflet.js with offline caching.[20] Map data from OpenStreetMap (OSM)[21] can be downloaded in advance, covering critical Hajj zones.

To balance performance and storage, map tiles can be compressed using MBTiles or GeoPackage formats. Overlays highlighting religious locations, pathways, and facilities can be preloaded and displayed interactively. Features like

[16] https://developers.google.com/maps
[17] https://www.openstreetmap.org
[18] https://osmand.net/
[19] https://docs.mapbox.com/ios/maps/overview/
[20] https://leafletjs.com
[21] https://www.openstreetmap.org

offline GPS tracking, point-of-interest search, and contextual tooltips further enhance usability in disconnected environments.

#### 2) MULTILINGUAL AI CHATBOT

This feature offers intelligent, multilingual assistance through a conversational interface capable of functioning offline. Designed with inclusivity in mind, the chatbot supports both text and speech-based interactions in languages such as Arabic, English, Urdu, Bahasa, French, and Turkish. It responds to queries related to navigation, ritual guidance, health and safety, and logistics. As shown in Figure 7, the chatbot facilitates natural interactions even in the absence of human guides or consistent network connectivity.

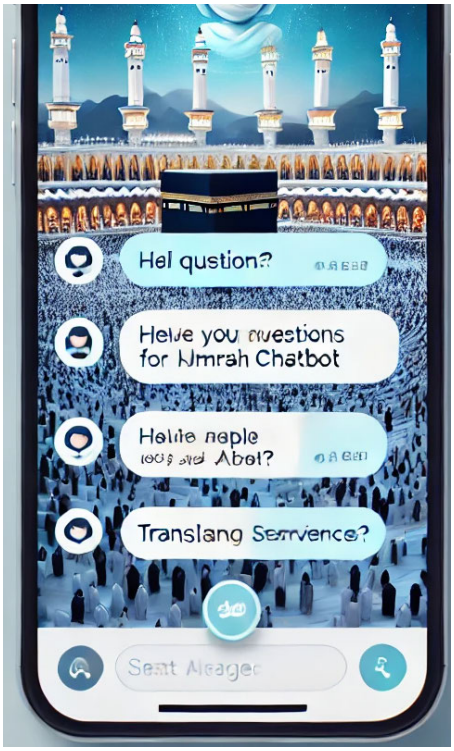



**FIGURE 7.** Multilingual AI chatbot interface.

##### *a: USE CASE*

When immediate help is unavailable, the chatbot can guide pilgrims by answering questions such as "Where is my group's tent?" or "How do I perform Rami al-Jamarat?" in the user's native language. This allows pilgrims to independently access location-specific or ritual-related information. Similar implementations have proven successful at large-scale events like Dubai Expo 2020 and in humanitarian settings, where multilingual AI assistants have been deployed to support information access in low-resource environments.[22]

[22]https://www.unhcr.org/innovation/chatbots-in-humanitarian-settings-revolutionary-a-fad-or-something-inbetween/

##### *b: IMPLEMENTATION CONSIDERATIONS*

The chatbot can be built using lightweight transformer-based NLP models such as DistilBERT [46] or ALBERT [47], optimized for low-latency and offline use. These models can be fine-tuned on domain-specific datasets related to Hajj rituals, geography, and health services. For multilingual support, text translation can be handled by models like M2M-100 [40], while voice input is processed using ASR systems such as Mozilla DeepSpeech [48]. Text-to-speech output can be generated using FastSpeech 2 [43] or cloud-based services like Google Text-to-Speech,[23] with fallback to offline options for disconnected scenarios. Integration with the offline guide map module allows the chatbot to deliver location-aware responses using preloaded GIS data. The interface supports language switching, voice commands, and real-time interaction, making it a comprehensive digital assistant tailored for diverse user needs during the Hajj pilgrimage.

### C. AUGMENTED REALITY (AR)-BASED SOLUTIONS

To address the navigational and situational challenges faced by pilgrims in crowded environments, the proposed system incorporates Augmented Reality (AR) technologies. AR enables users to interact with their physical surroundings through contextually relevant visual overlays—such as directional arrows, alerts, and labels—superimposed on real-world views. These solutions aim to provide intuitive, real-time guidance, enhance user awareness, and improve decision-making during the pilgrimage journey.

#### 1) PATHFINDER

The PathFinder feature leverages AR and crowd estimation to guide pilgrims through high-density areas safely and efficiently. Using a smartphone camera, the system overlays live video with dynamic, color-coded visual cues: red for high congestion, yellow for moderate density, and green for clear paths. This allows users to make informed movement decisions in real-time, especially during large-scale rituals such as Tawaf or the journey between Mina, Arafat, and Muzdalifah.

##### *a: USE CASE*

In dense areas like the Jamarat Bridge or Mataf, real-time crowd awareness can help prevent stampedes and reduce delays. For example, a pilgrim can hold up their phone to visualize congestion levels ahead and be guided toward safer, less crowded alternatives. Similar AR navigation has been successfully used in theme parks and large public venues to enhance crowd flow and safety [49].

##### *b: IMPLEMENTATION CONSIDERATIONS*

Crowd density estimation can be achieved using lightweight computer vision models such as CSRNet [50] or MCNN [51], which analyze live camera feeds to detect pedestrian density.

[23]https://cloud.google.com/text-to-speech

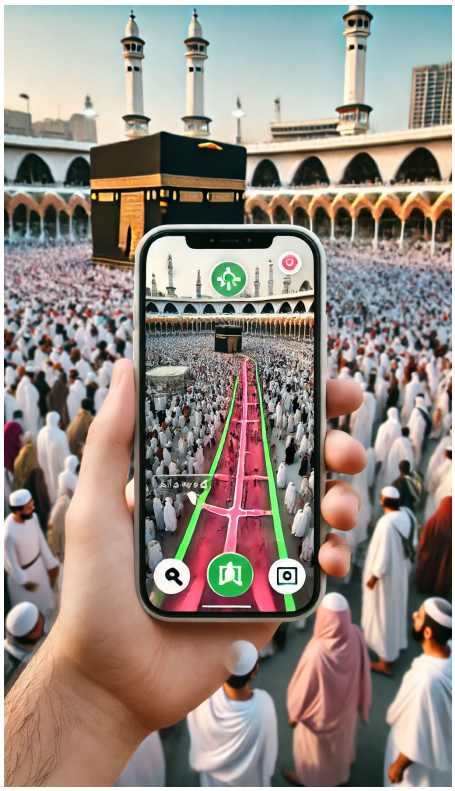


**FIGURE 8.** PathFinder AR interface for real-time navigation.

These outputs are transformed into visual heatmaps rendered via AR development frameworks like ARCore[24] (Android) or ARKit[25] (iOS). Color-coded overlays guide the user visually through congested areas. Integration with GPS and inertial sensors supports personalized path planning. The system can run efficiently using mobile inference runtimes like LiteRT[26] or Core ML,[27] allowing smooth operation even offline. This feature promotes safe navigation, supports emergency planning, and enhances mobility for users navigating complex and dynamic environments.

### 2) LOST PILGRIM TRACKER

The Lost Pilgrim Tracker is a safety-centric feature designed to help locate group members who have been unintentionally separated. Using Bluetooth Low Energy (BLE) technology for location approximation and AR for visual tracking, the system displays the approximate position of the missing pilgrim through a floating AR marker.

#### *a: USE CASE*

In high-mobility zones such as Mina tents or transit stations, elderly or vulnerable pilgrims may become separated from their group. Group members can activate the tracker, and through BLE signal detection, the system overlays the

[24]https://developers.google.com/ar
[25]https://developer.apple.com/augmented-reality/
[26]https://ai.google.dev/edge/litert
[27]https://developer.apple.com/machine-learning/core-ml/




**FIGURE 9.** Lost pilgrim tracker interface showing AR-guided search.

estimated direction and distance of the lost pilgrim on the camera view. This allows rapid reunion even in low-connectivity environments. Comparable technologies have been used at events and amusement parks to track children via wristbands and mobile apps [52].

#### *b: IMPLEMENTATION CONSIDERATIONS*

The system uses BLE beacons embedded in mobile devices to emit short-range encrypted signals. Nearby smartphones detect these signals and estimate distance based on Received Signal Strength Indicator (RSSI) values. The AR interface, built using ARKit or ARCore, renders a floating visual marker containing the pilgrim's photo and name. Sensor fusion—combining compass, gyroscope, and accelerometer data—improves spatial accuracy. Privacy is preserved through encrypted peer-to-peer BLE communication, and the system can optionally include a distress mode that automatically activates tracking after prolonged disconnection. Bluetooth Mesh Networking[28] ensures device-to-device communication in environments with poor or no internet access. This feature enhances group safety and enables faster reunification during emergencies or unintentional separation, especially for high-risk demographics.

### 3) MEETPOINT FINDER

The *MeetPoint Finder* is an intelligent AR-based navigation tool designed to assist pilgrims in locating predefined group meeting points—such as camps, buses, or gathering zones—with minimal confusion. By leveraging smartphone sensors and AR capabilities, the application overlays real-time digital cues onto the physical world, including directional arrows, distance meters, and proximity alerts through vibration or sound. This enables intuitive and efficient wayfinding in complex, visually homogeneous environments.

#### *a: USE CASE*

In regions like Mina, where tents appear nearly identical, pilgrims frequently face difficulty in locating their group's camp after rituals. The MeetPoint Finder enables users to visually follow AR overlays—such as ''120 meters ahead'' or

[28]https://www.bluetooth.com/learn-about-bluetooth/bluetooth-technology/bluetooth-mesh-networking/

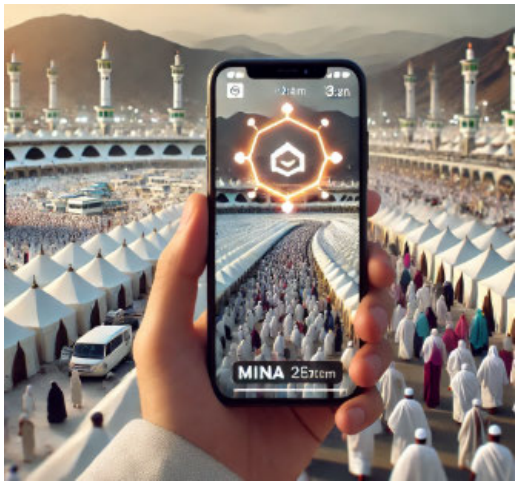



**FIGURE 10.** MeetPoint finder AR interface guiding users to saved destinations.

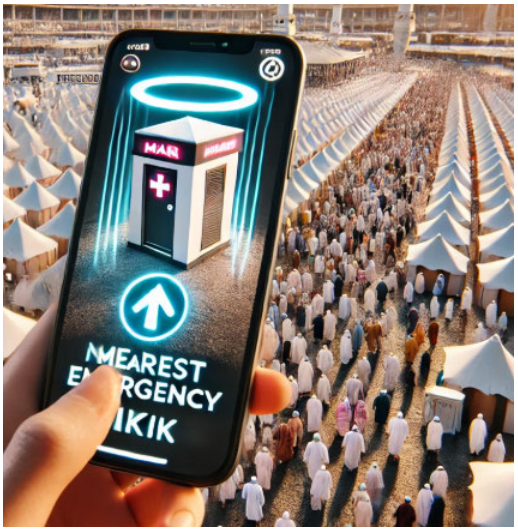



**FIGURE 11.** Emergency guide interface showing AR direction to nearby assistance.

''Turn left in 20 meters''—to reach a registered meet point, set earlier using GPS or shared through the group network. This approach draws on real-world deployments in large venues like airports and exhibitions. For instance, Gatwick Airport's AR wayfinding system has helped travelers find their gates using overlaid directions on live video feeds.[29] Similar use in pilgrimage environments enhances coordination, independence, and confidence.

#### b: IMPLEMENTATION CONSIDERATIONS

The system uses AR development platforms like Google ARCore[30] or Apple ARKit[31] to anchor virtual elements to the user's real-world view. Predefined meet points can be registered using GPS or on-site visual markers, stored locally or transmitted via secure Bluetooth or P2P connections. For real-time guidance, the app combines GPS, IMU sensors, and visual odometry. In GPS-limited settings like dense tent areas, Visual Positioning Systems (VPS)[32] can enhance accuracy by using surrounding landmarks. AR overlays include audio cues, distance tracking, and turn indicators—supporting diverse accessibility needs and ensuring usability in both connected and offline conditions.

### 4) EMERGENCY GUIDE

The *Emergency Guide* feature delivers real-time, AR-based direction to nearby emergency support locations—such as medical stations, safety kiosks, or evacuation routes—tailored for high-density and high-stress pilgrimage scenarios. By superimposing spatially accurate indicators over the user's camera view, the system ensures quick recognition and navigation, even in chaotic or unfamiliar environments.

#### a: USE CASE

Emergencies such as heatstroke, injury, or sudden illness may require fast response. A user witnessing such an event can activate the Emergency Guide, which immediately presents an on-screen path to the closest medical or emergency facility using directional arrows and distance counters. This concept aligns with real-world emergency navigation systems implemented in smart cities and disaster-prone areas. For example, AR has been used to direct civilians toward safety during earthquakes or building fires, improving evacuation outcomes [53].

#### b: IMPLEMENTATION CONSIDERATIONS

Emergency service locations are pre-tagged with precise GPS or VPS coordinates. Using ARKit or ARCore, the application can render live guidance markers—such as arrows or glowing icons—pointing toward these critical points. The system calculates the user's position through GPS and inertial sensors (accelerometer, gyroscope), combined with camera-based pose estimation for refined localization. Directional prompts are updated in real-time and supported by multilingual text, audio, and haptic feedback to ensure inclusive access. To maintain offline operability, emergency location data and AR assets can be cached locally. Bluetooth Low Energy (BLE) beacons or NFC tags placed at emergency points can further aid in proximity sensing and validation when network signals are weak or absent. This approach ensures fast, reliable assistance in emergency contexts, enhancing safety and user confidence.

### 5) VIRTUAL GUIDE TO HOLY SITES

The *Virtual Guide to Holy Sites* is an immersive, AR-enabled feature designed to familiarize pilgrims with significant religious locations associated with Hajj and Umrah before they arrive. Through interactive walkthroughs and realistic 3D simulations, users can virtually explore key landmarks—

[29] https://interestingengineering.com/culture/gatwick-airport-uses-augmented-reality-help-catching-flights
[30] https://developers.google.com/ar
[31] https://developer.apple.com/augmented-reality/
[32] https://developers.google.com/ar/data-types#visual-positioning-system

such as the Kaaba, Safa and Marwah, the Jamarat Bridge, and Mount Arafat—enhancing spatial awareness and ritual understanding. This feature is especially beneficial for first-time, elderly, or anxious pilgrims, supporting mental preparedness and confidence.

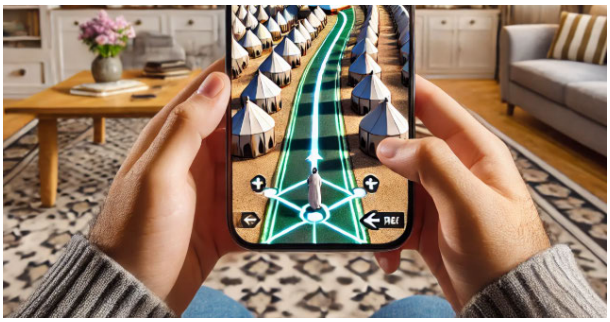


**FIGURE 12.** Virtual Guide interface previewing holy sites in AR.

#### a: USE CASE

Pilgrims unfamiliar with Hajj procedures may feel overwhelmed by the scale and logistics of rituals such as Sa'i between Safa and Marwah or standing at Arafat. Using this feature, users can preview routes, entrances, and landmarks virtually—either at home or while traveling—building familiarity in advance. AR-based educational systems have been applied in heritage tourism and cultural education. For instance, AR tools have enabled users to remotely engage with sacred architecture, promoting both emotional and cognitive preparedness [54].

#### b: IMPLEMENTATION CONSIDERATIONS

This system can be developed using Unity with AR Foundation[33] or Unreal Engine integrated with ARKit or ARCore. Spatially accurate 3D environments can be created from photogrammetry, LiDAR scans, or GIS datasets and optimized for real-time rendering on mobile devices. Features include multilingual narration, annotated walkthroughs, ritual instruction overlays, and user interaction via touch or device motion. Enhancements such as light estimation, occlusion handling, and surface detection contribute to realism and user immersion. By simulating key rituals and spaces, the Virtual Guide promotes a more informed and confident pilgrimage experience.

### 6) SCAN AND UNDERSTAND

The *Scan and Understand* feature integrates Optical Character Recognition (OCR) and Augmented Reality (AR) to provide real-time translation of written content within a live camera feed. By pointing their smartphone at signage, notices, or printed materials, users can instantly view translated text superimposed over the original content—without switching apps or losing context.

[33]https://docs.unity3d.com/Packages/com.unity.xr.arfoundation




**FIGURE 13.** Scan and understand interface overlaying translated AR text.

#### a: USE CASE

During Hajj and Umrah, essential information is often presented only in Arabic. Non-Arabic speakers can struggle to interpret such messages. With this feature, users can view real-time translations (e.g., "Sortie de secours" in French) overlaid directly on the original sign. This facilitates safer and more independent navigation in multilingual, high-density environments. Comparable systems like Google Lens[34] and Apple Live Text[35] have demonstrated the value of live OCR and translation in everyday travel and accessibility contexts.

#### b: IMPLEMENTATION CONSIDERATIONS

The system combines OCR engines such as Tesseract OCR[36] or Google Cloud Vision OCR[37] with Neural Machine Translation (NMT) models like M2M-100 [40] or T5 [39]. These tools enable accurate recognition and translation of multilingual printed text in real time. Using AR platforms like ARKit[38] and ARCore,[39] the translated output is anchored over the original content, maintaining visual consistency. Additional capabilities include auto language detection, font matching, and support for poor lighting conditions. To ensure offline usability, components can run locally using LiteRT [40] for OCR and translation, with optional cloud fallback for higher accuracy. Extended features may include voice playback of translations, pronunciation guides, and pop-up definitions—offering a comprehensive, real-time language support system tailored to pilgrims' diverse linguistic needs.

## D. ADDITIONAL SUPPORT SERVICES

To further augment the technological ecosystem proposed for supporting pilgrims, the system integrates supplementary services that can also address critical needs. These features

[34]https://lens.google
[35]https://support.apple.com/guide/iphone/use-live-text-iphcf0b71b0e/ios
[36]https://github.com/tesseract-ocr/tesseract
[37]https://cloud.google.com/vision
[38]https://developer.apple.com/augmented-reality/
[39]https://developers.google.com/ar
[40]https://ai.google.dev/edge/litert

aim to promote well-being, safety, and convenience during the pilgrimage.

#### 1) CHARITY INFORMATION AND LOCATIONS

This feature enables pilgrims to locate and access real-time information about nearby charity centers through an interactive map interface powered by Geographic Information Systems (GIS). It enhances the accessibility of essential services—such as food, water, medical aid, and camp—particularly in high-density or resource-limited areas like Mina, Arafat, and Muzdalifah. The dynamic interface presents locations of nearby charitable centers enriched with metadata, including organization names (e.g., *X Charity*, *Y Welfare Association*), types of services provided, categories of food available (e.g., rice, bread, dates), and operating schedules. Turn-by-turn navigation assists users in efficiently reaching selected centers from their current location.

##### *a: USE CASE*

During Hajj, quick and equitable access to food and medical services is vital—particularly for elderly or vulnerable pilgrims. A user seeking a hot meal or first aid can filter services via the app and receive visual directions to the nearest charity center, eliminating reliance on verbal inquiries or language-specific instructions. GIS-enabled humanitarian coordination has demonstrated significant benefits in mass gatherings by improving response time and equitable resource allocation [55], [56]. Applying these techniques in a pilgrimage setting ensures effective outreach to those most in need.




**FIGURE 14.** Charity information interface.

##### *b: IMPLEMENTATION CONSIDERATIONS*

The frontend can be developed using cross-platform frameworks like Flutter or React Native. Map interfaces are rendered through SDKs such as Mapbox[41] or Google Maps.[42] The user's GPS coordinates are leveraged for real-time positioning and routing. The backend utilizes geospatially indexed databases (e.g., MongoDB, Firebase) to store and serve charity data. RESTful APIs enable continuous data exchange, while cloud messaging services support live updates. Navigation is supported through services like the Mapbox Directions API.[43] Data sources include partnerships with verified nonprofit organizations, moderated user submissions, and OpenStreetMap.[44] Privacy safeguards—such as encrypted data transmission and user consent for GPS access—are incorporated into the architecture.

#### 2) TRANSPORTATION AND NAVIGATION

This feature integrates route optimization, live GPS tracking, and real-time transit updates to support efficient movement during key phases of the pilgrimage. It can comprise two submodules: Tafweej Bus Schedules, which provides real-time information on bus availability, capacity alerts, and scheduling to facilitate mass group transport; and Masaraat Navigation, which delivers personalized pedestrian routing that adapts to dynamic crowd densities and environmental conditions. Together, these components support safe and coordinated navigation between major pilgrimage sites such as the Haram, Mina, Arafat, and Jamarat.

##### *a: USE CASE*

During large-scale transitions—e.g., from Arafat to Muzdalifah—millions of pilgrims must travel simultaneously. This feature allows pilgrims to view bus schedules, avoid overcrowded routes, and follow adaptive walking paths with GPS- or AR-enabled navigation. It reduces confusion and enhances safety during high-risk movements. These smart mobility strategies are modeled on implementations at global events such as the Beijing Olympics and Shanghai Expo [57]. Studies have also explored their relevance in Mecca's context for improving ritual logistics [58].

##### *b: IMPLEMENTATION CONSIDERATIONS*

The frontend supports mobile development using Flutter or Kotlin Multiplatform. Navigation features are delivered using SDKs such as Mapbox Navigation[45] or HERE Maps.[46] GPS and background services track user movement for real-time route guidance. The backend incorporates Firebase Cloud Messaging[47] or MQTT for real-time communication. Bus route data is sourced via APIs like Google Transit,[48] while routing algorithms (e.g., Dijkstra, A*) compute the shortest paths dynamically. Geospatial analytics and geofencing are used to adapt routes in real-time based on congestion or environmental constraints. Scalable cloud infrastructure (e.g., AWS or Google Cloud) ensures performance during peak

[41] https://docs.mapbox.com/android/maps/overview/
[42] https://developers.google.com/maps
[43] https://docs.mapbox.com/api/navigation/directions/
[44] https://www.openstreetmap.org/about
[45] https://docs.mapbox.com/android/navigation/overview/
[46] https://developer.here.com/
[47] https://firebase.google.com/docs/cloud-messaging
[48] https://developers.google.com/transit




**FIGURE 15.** Transportation and navigation interface.




**FIGURE 16.** Haramain explorer interface.

load. This modular design ensures robustness, adaptability, and responsiveness, providing context-aware mobility guidance under the extreme logistical demands of pilgrimage events.

### 3) HARAMAIN EXPLORER

The Haramain Explorer is an interactive feature that offers pilgrims easy access to essential services and culturally significant locations across pilgrimage zones. By combining geospatial visualization with a categorized directory of key resources, this module enhances situational awareness, decision-making, and the overall pilgrimage experience. Through an intuitive mobile interface, users can explore a variety of points of interest—including historical landmarks, healthcare centers, police departments, markets, food courts, public parks, and telecommunication facilities. Each location entry includes detailed metadata, such as service descriptions, images, operational hours, and navigation options. Notably, the module supports both spiritual and logistical needs. Pilgrims can locate and learn about religious sites associated with Hajj and Umrah, as well as identify nearby emergency services or public amenities. In the dense urban settings of Makkah and Madinah, timely access to accurate location-based information improves navigation, safety, and overall experience, as supported by related studies [59], [60].

#### *a: USE CASE*

In dense environments such as Mina or the Grand Masjid surroundings, pilgrims may struggle to find critical services. For example, a user looking for the Cave of Hira or the Quba Mosque can use the Haramain Explorer to receive multimedia information and real-time navigation. In emergency scenarios, the app provides instant routing to the nearest clinic, reducing dependency on verbal inquiries or staff assistance. Additionally, pilgrims can identify commercial amenities like food courts and telecom points to manage their daily needs more efficiently. This use aligns with smart tourism and smart city paradigms, where geospatial systems and mobile services are central to managing visitor flows, ensuring safety, and enhancing user satisfaction [59].

#### *b: IMPLEMENTATION CONSIDERATIONS*

The Haramain Explorer leverages a GIS-enabled service-oriented architecture combining real-time spatial data with a dynamic, mobile-friendly frontend. The interface is developed using frameworks such as Flutter or React Native and incorporates geospatial SDKs like Mapbox[49] or Leaflet[50] for map rendering, service categorization, and interactive routing.

A cloud-based backend—powered by Firebase Firestore or MongoDB—stores geotagged entries for various facilities, enriched with metadata such as name, category, location, hours of operation, and contact information. RESTful APIs facilitate seamless data exchange, while webhook-based triggers enable dynamic updates when new services are

[49]https://docs.mapbox.com
[50]https://leafletjs.com

**TABLE 5.** Summarized implementation considerations.

| Functionality | Implementation Considerations with Weblinks |
|---|---|
| Text-to-Text Translation | T5 (Google), M2M-100 (Facebook), LiteRT, ONNX Runtime, Google Cloud Translation API, Azure Translator |
| Voice-to-Text Translation | Google Speech-to-Text, Mozilla DeepSpeech, T5, M2M-100, ONNX Runtime, LiteRT |
| Text-to-Voice Translation | Tacotron 2, FastSpeech 2, Google Cloud TTS, Amazon Polly, ONNX Runtime, LiteRT |
| Text/Voice to Image/Map | BERT, T5, Google Speech-to-Text, DeepSpeech, Google Maps API, OpenStreetMap |
| Offline Guide Maps | Mapbox GL Native, Leaflet, OpenStreetMap, MBTiles, GeoPackage |
| Multilingual AI ChatBot | DistilBERT, ALBERT, M2M-100, DeepSpeech, FastSpeech 2, Google Cloud TTS, OpenStreetMap |
| PathFinder | CSRNet, MCNN, ARCore, ARKit, LiteRT, Core ML |
| Lost Pilgrim Tracker | Bluetooth Mesh, ARCore, ARKit |
| MeetPoint Finder | ARCore, ARKit, Visual Positioning System (VPS) |
| Emergency Guide | ARCore, ARKit, Visual Positioning System (VPS) |
| Virtual Guide to Holy Sites | Unity AR Foundation, Unreal Engine, ARKit, ARCore |
| Scan and Understand | Tesseract OCR, Google Cloud Vision OCR, M2M-100, T5, ARCore, ARKit, LiteRT |
| Charity Information and Locations | Flutter, React Native, Mapbox, Google Maps, Firebase, MongoDB, Mapbox Directions API |
| Transportation and Navigation | Flutter, Kotlin Multiplatform, Mapbox Navigation, HERE Maps, Firebase Cloud Messaging, Google Transit API |
| Haramain Explorer | Flutter, React Native, Mapbox, Leaflet, Firebase Firestore, MongoDB, OpenStreetMap, GeoFire, PostGIS |

added. To enhance spatial intelligence, indoor mapping and advanced querying can be implemented using tools like PostGIS or Firebase GeoFire.[51] For external mapping data, integration with OpenStreetMap[52] ensures flexibility and extensibility. Sensitive data access—such as locations of police departments—can be safeguarded through role-based access control and encrypted communication protocols.

## V. RESULTS

This section presents the results of a our study conducted to evaluate and prioritize smart mobile features for Hajj pilgrims as described in Section III. The study included nine domain experts who rated the proposed features in two iterative rounds using a 5-point Likert scale. The process was structured in three main phases: (1) identifying features for inclusion based on *challenge relevance*, and (2) evaluating included features across four feasibility dimensions to derive a final prioritization and (3) feature prioritization.

### A. FEATURE INCLUSION BASED ON CHALLENGE RELEVANCE

In the first phase, each feature was evaluated on its relevance to the challenges faced by pilgrims during Hajj. This was defined as the degree to which the feature addressed one or more challenges identified in Section III-B. A feature was considered to have achieved *consensus for inclusion* if the median rating in Round 2 was $\geq$ 4.0. While all features reached the consensus threshold of 4.0 or higher in Round 1, a second Delphi round was still conducted to strengthen the reliability of expert judgments and confirm consistency in perceptions. The second round offered participants the opportunity to reconsider their ratings after reviewing group feedback and aggregated statistics from Round 1. This iterative feedback mechanism is a key element of the Delphi method, promoting reflection and convergence towards stable group consensus.

Table 6 presents the median scores from both Delphi rounds (derived from Table 12 and Table 13). All features maintained or improved their median scores in Round 2, with all features receiving a median of 4.0 or 5.0, indicating sustained and strong agreement on their relevance. Notably, the *Text/Voice to Image or Map* feature showed an increase in median score between rounds, as indicated by a red upward arrow (↑), reflecting growing consensus. Green-shaded cells denote features that met or exceeded the consensus threshold in both rounds.

**TABLE 6.** Challenge relevance ratings.

| Feature | Median R1 | Median R2 |
|---|---|---|
| Text-to-Text Translation | **4.0** | **4.0** |
| Voice-to-Text Translation | **5.0** | **5.0** |
| Text-to-Voice Translation | **4.0** | **4.0** |
| Text/Voice to Image or Map | **4.0** | ↑ **5.0** |
| Offline Guide Maps | **5.0** | **5.0** |
| Multilingual AI ChatBot | **5.0** | **5.0** |
| PathFinder | **5.0** | **5.0** |
| LostPilgrim Tracker | **5.0** | **5.0** |
| MeetPoint Finder | **4.0** | **4.0** |
| EmergencyGuidance | **5.0** | **5.0** |
| Virtual Guide to Holy Sites | **5.0** | **5.0** |
| Scan and Understand | **5.0** | **5.0** |
| Charity Info & Locations | **4.0** | **4.0** |
| Transportation & Navigation | **5.0** | **5.0** |
| Haramain Explorer | **5.0** | **5.0** |

### B. FEASIBILITY EVALUATION ACROSS FOUR DIMENSIONS

Following the consensus-based inclusion of the proposed features, a second phase of expert evaluation was conducted to assess their feasibility across four critical dimensions. These included: **(i) Infrastructure Feasibility (IF)** — Assesses how realistically a feature can be deployed using in-house or on-premise infrastructure, without relying on third-party platforms. This ensures full data control, reduces external dependencies, and mitigates security and privacy

[51] https://github.com/firebase/geofire-js
[52] https://www.openstreetmap.org/about

risks. **(ii) Development Feasibility (DF)** — Assesses the ease and speed of feature development in terms of time, resource availability, and cost-efficiency using available technologies and internal capabilities. **(iii) Regulatory Feasibility (RF)** — Evaluates how well a feature complies with legal, and administrative protocols, particularly those set by the Saudi government for Hajj operations. **(iv) Environmental Feasibility (EF)** — Considers the robustness and effectiveness of each feature under real-world Hajj conditions, including extreme heat, crowd congestion, and unreliable network connectivity.

Each feature was rated by a panel of nine domain experts using a 5-point likert scale over two Delphi rounds. Experts assigned ratings based on the criteria defined in Section III-C. To improve the reliability and consensus of the evaluation, a second Delphi round was conducted. Experts whose initial ratings deviated significantly from the aggregated group response were shown the mean and median values from Round 1 and asked to reconsider their assessments. This feedback process facilitated convergence and refinement of expert opinion.

Mean values from Round 2 were used to guide feature prioritization. Tables 7 through 10 summarize the feasibility scores across the four dimensions. Notably, features such as *Text-to-Voice Translation*, *Lost Pilgrim Tracker*, and *Multilingual AI ChatBot* showed an upward shift in feasibility perception between rounds. These improvements are marked with a red upward arrow (↑) preceding the updated Round 2 values.

**TABLE 7.** Infrastructure feasibility – mean ratings (↑ before value).

| Feature | Mean R1 | Mean R2 |
|---|---|---|
| Text-to-Text Translation | 4.33 | 4.33 |
| Voice-to-Text Translation | 3.89 | 3.89 |
| Text-to-Voice Translation | 4.00 | 4.00 |
| Text or Voice to Image or Map | 3.22 | ↑ 3.33 |
| Offline Guide Maps | 4.00 | 4.00 |
| Multilingual AI ChatBot | 3.33 | 3.33 |
| PathFinder | 3.33 | 3.33 |
| LostPilgrim Tracker | 3.33 | 3.33 |
| MeetPoint Finder | 3.78 | 3.78 |
| EmergencyGuide | 4.00 | 4.00 |
| Virtual Guide to Holy Sites | 3.67 | 3.67 |
| Scan and Understand | 3.67 | 3.67 |
| Charity Information and Locations | 4.44 | 4.44 |
| Transportation and Navigation | 3.89 | 3.89 |
| Haramain Explorer | 3.67 | ↑ 3.78 |

As shown in Table 7 (derived from Table 14 and Table 15), features such as *Charity Info & Locations* and *Transportation & Navigation* scored the highest in infrastructure feasibility, indicating they are well-suited for deployment using on-premise infrastructure. Modest increases were observed for features like *Text or Voice to Image or Map* and *Multilingual AI ChatBot*, reflecting improved expert agreement in Round 2.

As shown in Table 8, (derived from Table 16 and Table 17), Development feasibility showed particularly strong ratings for *Text-to-Text Translation* and *Text-to-Voice Translation*. Several features, including *EmergencyGuide* and *Virtual Guide to Holy Sites*, exhibited increased ratings, reflecting a growing perception of their development viability.

**TABLE 8.** Development feasibility – mean ratings with increases highlighted (↑ before value).

| Feature | Mean R1 | Mean R2 |
|---|---|---|
| Text-to-Text Translation | 4.56 | ↑ 4.67 |
| Voice-to-Text Translation | 4.22 | 4.22 |
| Text-to-Voice Translation | 4.33 | 4.33 |
| Text or Voice to Image or Map | 3.67 | 3.67 |
| Offline Guide Maps | 3.89 | 3.89 |
| Multilingual AI ChatBot | 3.67 | 3.67 |
| PathFinder | 3.56 | 3.56 |
| LostPilgrim Tracker | 3.22 | ↑ 3.33 |
| MeetPoint Finder | 3.56 | ↑ 3.67 |
| EmergencyGuide | 3.67 | ↑ 3.78 |
| Virtual Guide to Holy Sites | 3.67 | ↑ 3.78 |
| Scan and Understand | 3.56 | ↑ 3.67 |
| Charity Info & Locations | 3.89 | ↑ 4.00 |
| Transportation & Navigation | 3.56 | ↑ 3.78 |
| Haramain Explorer | 3.33 | ↑ 3.56 |

**TABLE 9.** Regulatory feasibility – mean ratings with increases highlighted (↑ before value).

| Feature | Mean R1 | Mean R2 |
|---|---|---|
| Text-to-Text Translation | 3.89 | ↑ 4.33 |
| Voice-to-Text Translation | 3.56 | ↑ 4.11 |
| Text-to-Voice Translation | 3.89 | ↑ 4.44 |
| Text or Voice to Image or Map | 3.33 | ↑ 3.78 |
| Offline Guide Maps | 4.00 | ↑ 4.44 |
| Multilingual AI ChatBot | 3.78 | ↑ 4.22 |
| PathFinder | 3.67 | ↑ 4.00 |
| LostPilgrim Tracker | 3.67 | ↑ 3.89 |
| MeetPoint Finder | 3.67 | ↑ 3.89 |
| EmergencyGuide | 3.67 | ↑ 4.11 |
| Virtual Guide to Holy Sites | 3.67 | ↑ 3.89 |
| Scan and Understand | 3.89 | ↑ 4.11 |
| Charity Info & Locations | 3.78 | ↑ 4.22 |
| Transportation & Navigation | 3.67 | ↑ 4.22 |
| Haramain Explorer | 3.67 | ↑ 3.89 |

Table 9 (derived from Table 18 and Table 19) highlights widespread improvement in perceptions of regulatory compliance. Notably, features such as *Text-to-Voice Translation*, *Offline Guide Maps*, and *Multilingual AI ChatBot* experienced marked gains, suggesting increasing alignment with Islamic and local regulations.

**TABLE 10.** Environmental feasibility – mean ratings with increases highlighted (↑ before value).

| Feature | Mean R1 | Mean R2 |
|---|---|---|
| Text-to-Text Translation | 3.22 | ↑ 3.33 |
| Voice-to-Text Translation | 3.44 | ↑ 3.78 |
| Text-to-Voice Translation | 3.33 | ↑ 3.56 |
| Text or Voice to Image or Map | 3.22 | ↑ 3.33 |
| Offline Guide Maps | 3.67 | ↑ 3.89 |
| Multilingual AI ChatBot | 3.33 | ↑ 3.67 |
| PathFinder | 3.67 | ↑ 3.89 |
| LostPilgrim Tracker | 3.78 | 3.78 |
| MeetPoint Finder | 3.67 | ↑ 3.78 |
| EmergencyGuide | 4.00 | 4.00 |
| Virtual Guide to Holy Sites | 3.56 | ↑ 3.67 |
| Scan and Understand | 3.22 | ↑ 3.33 |
| Charity Info & Locations | 3.89 | ↑ 4.11 |
| Transportation & Navigation | 3.56 | 3.56 |
| Haramain Explorer | 4.33 | 4.33 |

Environmental feasibility results in Table 10 (derived from Table 20 and Table 21) demonstrate strong performance by features like *EmergencyGuide* and *Haramain Explorer*, which are perceived as resilient in real-world Hajj conditions. Modest improvements were seen in features requiring better environmental adaptability, such as *Voice-to-Text* and *Text-to-Voice* tools.

### C. FEATURE PRIORITIZATION USING WEIGHTED FEASIBILITY SCORES

To derive a final prioritization of the proposed features, a weighted scoring model was applied based on the four feasibility dimensions evaluated in our study. The proposed formula (Equation 1) combines the Round 2 mean scores for each dimension, reflecting aggregated ratings of nine domain experts.

$$\text{Priority Score}_i = 0.30 \cdot \text{IF}_i + 0.20 \cdot \text{DF}_i + 0.30 \cdot \text{RF}_i + 0.20 \cdot \text{EF}_i \quad (1)$$

Greater weight was assigned to *Infrastructure Feasibility* and *Regulatory Feasibility* (30% each), reflecting their critical importance in ensuring secure, in-house deployment and alignment with Saudi legal protocols. These dimensions are also widely recognized in the literature as primary barriers to implementing digital solutions at scale in complex event environments. For instance, research on smart-city governance in developing countries underlines that sustainable digital transformation can only be achieved alongside infrastructure investment and regulatory reform [61]. Similarly, studies in the context of Hajj emphasize that infrastructural readiness—such as network coverage, system reliability, and interoperability—is crucial for successful technology deployment during pilgrimage [3]. The weighting scheme was therefore informed by the authors' expertise and aligned with these findings. While *Development Feasibility* and *Environmental Feasibility* were also considered important, they were assigned slightly lower weights (20% each) to reflect their secondary—but still relevant—impact on deployment viability.

Table 11 presents the resulting priority scores, with features ranked from highest to lowest. The top-ranked feature was *Charity Info & Locations* (score = 4.220), indicating its strong technical feasibility, low regulatory risk, and practical utility during pilgrimage. *Text-to-Text Translation* (4.198), *Text-to-Voice Translation* (4.110), and *Offline Guide Maps* (4.088) also ranked highly, suggesting that multilingual communication and offline navigational aids are both viable and impactful. At the mid-range, features like *Transportation & Navigation* (3.901), *Haramain Explorer* (3.879), and *MeetPoint Finder* (3.791) showed balanced feasibility but slightly lower scores in regulatory or infrastructure categories. The lowest-ranked feature was *Text or Voice to Image or Map* (3.533), which received lower feasibility ratings across several dimensions, particularly infrastructure and environmental adaptability. Similarly, features like *LostPilgrim Tracker* (3.588) and *PathFinder* (3.689) scored lower, possibly due to their dependence on real-time AR and BLE technologies that pose deployment challenges in congested or offline settings.

**TABLE 11. Feature prioritization based on weighted feasibility scores.**

| Feature | Priority Score |
|---|---|
| Charity Info & Locations | 4.220 |
| Text-to-Text Translation | 4.198 |
| Text-to-Voice Translation | 4.110 |
| Offline Guide Maps | 4.088 |
| Voice-to-Text Translation | 4.000 |
| EmergencyGuide | 3.989 |
| Transportation & Navigation | 3.901 |
| Haramain Explorer | 3.879 |
| MeetPoint Finder | 3.791 |
| Virtual Guide to Holy Sites | 3.758 |
| Scan and Understand | 3.734 |
| Multilingual AI ChatBot | 3.733 |
| PathFinder | 3.689 |
| LostPilgrim Tracker | 3.588 |
| Text or Voice to Image or Map | 3.533 |

## VI. DISCUSSION

This section discusses the meaning of the results, and explores the implications for system designers, developers, and policy stakeholders. It also highlights study limitations and proposes avenues for future research.

### A. INTERPRETATION OF RESULTS

Our study revealed strong expert consensus on the inclusion of all the proposed features, each addressing challenges encountered by pilgrims during Hajj. Features such as *Charity Info & Locations* (4.256), *Text-to-Text Translation* (4.198), and *Text-to-Voice Translation* (4.110) received the highest scores, reflecting their practical feasibility and strong alignment with pressing user needs such as access to reliable information (CH3), communication across language barriers (CH1), and locating aid services (CH8).

The prioritization of translation-related features reinforces the importance of mitigating linguistic barriers—an issue consistently reported in Hajj studies [62], [63]. Notably, all three translation modes (Text-to-Text, Voice-to-Text and Text-to-Voice Translation) ranked among the top seven features, indicating a shared expert belief that real-time, multilingual communication tools are foundational to improving pilgrim autonomy and safety. This is supported by Ahmad et al. [64], who advocate for context-aware, language-sensitive applications for religious events. Navigation and offline support features also ranked highly. For instance, *Offline Guide Maps* (4.088) and *Transportation & Navigation* (4.066) address navigational confusion (CH2) and transport-related uncertainty (CH8), while also catering to digital access issues in low-connectivity zones (CH7). Their high scores suggest feasibility even under environmental or infrastructural constraints, a known challenge during peak Hajj days. Emergency support tools like the *Emergency Guide* (4.022) also ranked well, directly tackling safety-related challenges (CH5). Meanwhile, mid-ranked features such as *Voice-to-Text Translation* (4.000)

**TABLE 12.** Challenge relevance - round 1 ratings.

| Participant | Text-to-Text Translation | Voice-to-Text Translation | Text-to-Voice Translation | Text or Voice to Image or Map | Offline Guide Maps | Multilingual AI ChatBot | PathFinder | LostPilgrim Tracker | MeetPoint Finder | Emergency Guidance | Virtual Guide to Holy Sites | Scan and Understand | Charity Info & Locations | Transportation & Navigation | Haramain Explorer |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| P1 | 3 | 4 | 3 | 5 | 4 | 5 | 5 | 4 | 5 | 5 | 5 | 5 | 3 | 5 | 5 |
| P2 | 2 | 3 | 4 | 4 | 5 | 3 | 4 | 5 | 4 | 4 | 3 | 3 | 4 | 5 | 3 |
| P3 | 4 | 5 | 5 | 5 | 4 | 5 | 5 | 5 | 4 | 5 | 5 | 5 | 4 | 5 | 5 |
| P4 | 5 | 5 | 5 | 5 | 5 | 5 | 5 | 5 | 5 | 5 | 5 | 5 | 5 | 5 | 5 |
| P5 | 5 | 4 | 2 | 4 | 5 | 5 | 4 | 3 | 3 | 3 | 4 | 4 | 2 | 3 | 3 |
| P6 | 3 | 2 | 2 | 3 | 5 | 4 | 4 | 4 | 4 | 4 | 5 | 5 | 4 | 4 | 4 |
| P7 | 5 | 5 | 5 | 5 | 5 | 5 | 5 | 5 | 5 | 5 | 5 | 5 | 5 | 5 | 5 |
| P8 | 5 | 5 | 5 | 4 | 4 | 4 | 4 | 4 | 5 | 5 | 4 | 5 | 5 | 5 | 5 |
| P9 | 3 | 5 | 3 | 3 | 4 | 5 | 5 | 5 | 2 | 3 | 4 | 2 | 2 | 5 | 5 |
| **Median** | 4 | 5 | 4 | 4 | 5 | 5 | 5 | 5 | 4 | 5 | 5 | 5 | 4 | 5 | 5 |

**TABLE 13.** Challenge relevance – round 2 ratings.

| Participant | Text-to-Text Translation | Voice-to-Text Translation | Text-to-Voice Translation | Text or Voice to Image or Map | Offline Guide Maps | Multilingual AI ChatBot | PathFinder | LostPilgrim Tracker | MeetPoint Finder | Emergency Guidance | Virtual Guide to Holy Sites | Scan and Understand | Charity Info & Locations | Transportation & Navigation | Haramain Explorer |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| P1 | 3 | 4 | 3 | 5 | 4 | 5 | 5 | 4 | 5 | 5 | 5 | 5 | 3 | 5 | 5 |
| P2 | 2 | 3 | 4 | 4 | 5 | 3 | 4 | 5 | 4 | 4 | 3 | 3 | 4 | 5 | 3 |
| P3 | 4 | 5 | 5 | 5 | 4 | 5 | 5 | 5 | 4 | 5 | 5 | 5 | 4 | 5 | 5 |
| P4 | 5 | 5 | 5 | 5 | 5 | 5 | 5 | 5 | 5 | 5 | 5 | 5 | 5 | 5 | 5 |
| P5 | 5 | 4 | 2 | 4 | 5 | 5 | 4 | 3 | 3 | 3 | 4 | 4 | 2 | 3 | 3 |
| P6 | 3 | 3 | 3 | 3 | 5 | 4 | 4 | 4 | 4 | 4 | 5 | 5 | 4 | 4 | 4 |
| P7 | 5 | 5 | 5 | 5 | 5 | 5 | 5 | 5 | 5 | 5 | 5 | 5 | 5 | 5 | 5 |
| P8 | 5 | 5 | 5 | 5 | 4 | 5 | 4 | 5 | 5 | 5 | 5 | 5 | 5 | 5 | 5 |
| P9 | 3 | 5 | 3 | 3 | 4 | 5 | 5 | 5 | 2 | 3 | 4 | 2 | 2 | 5 | 5 |
| **Median** | 4 | 5 | 4 | 5 | 5 | 5 | 5 | 5 | 4 | 5 | 5 | 5 | 4 | 5 | 5 |

and *Haramain Explorer* (3.879) offer a mix of situational utility and feasible implementation, albeit with slightly lower infrastructural readiness or regulatory clarity. Lower-priority features, such as *Text/Voice to Image or Map Translation* (3.533), ranked lowest due to lower ratings across feasibility dimensions—particularly infrastructure and environmental robustness. While the feature conceptually addresses accessibility for illiterate users or those with limited map-reading skills (CH9), its complexity and technological dependency may hinder deployment under current conditions, echoing concerns raised by Hashim et al. [65] regarding overly sophisticated interfaces in high-stress environments. Finally, the comparison (Table 2) indicates that Nusuk currently supports permits, booking, and basic guidance. The features which are identified by our expert panel—particularly AI-driven multilingual assistance, AR-based safety/navigation, and structured emergency workflows—are not explicitly documented (see also Section III-B). This supports the novelty and practical value of our identified feature set. Overall, the findings confirm that high-priority features tend to be those that are not only relevant to core pilgrim needs but also practical within existing Hajj infrastructure and regulatory boundaries. These insights provide a strong foundation for selecting features that are both impactful and deployable in the short to medium term.

**TABLE 14.** Infrastructure feasibility – round 1 ratings.

| Participant | Text-to-Text Translation | Voice-to-Text Translation | Text-to-Voice Translation | Text or Voice to Image or Map | Offline Guide Maps | Multilingual AI ChatBot | PathFinder | LostPilgrim Tracker | MeetPoint Finder | Emergency Guidance | Virtual Guide to Holy Sites | Scan and Understand | Charity Info & Locations | Transportation & Navigation | Haramain Explorer |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| P1 | 4 | 3 | 3 | 2 | 3 | 1 | 2 | 2 | 2 | 3 | 3 | 4 | 4 | 2 | 4 |
| P2 | 4 | 3 | 4 | 1 | 3 | 2 | 3 | 3 | 3 | 4 | 3 | 2 | 4 | 3 | 4 |
| P3 | 4 | 4 | 4 | 3 | 4 | 2 | 3 | 3 | 3 | 3 | 4 | 3 | 4 | 3 | 3 |
| P4 | 5 | 5 | 5 | 5 | 5 | 5 | 5 | 5 | 5 | 5 | 5 | 5 | 5 | 5 | 5 |
| P5 | 5 | 5 | 5 | 4 | 4 | 4 | 4 | 4 | 5 | 5 | 4 | 4 | 5 | 5 | 5 |
| P6 | 3 | 3 | 3 | 3 | 3 | 3 | 3 | 3 | 3 | 3 | 3 | 3 | 3 | 3 | 2 |
| P7 | 5 | 5 | 5 | 5 | 5 | 4 | 4 | 4 | 4 | 4 | 4 | 4 | 5 | 5 | 4 |
| P8 | 5 | 5 | 5 | 4 | 5 | 5 | 3 | 3 | 5 | 5 | 3 | 5 | 5 | 4 | 3 |
| P9 | 4 | 2 | 2 | 2 | 4 | 4 | 3 | 3 | 4 | 4 | 4 | 3 | 5 | 5 | 3 |
| **Mean** | 4.33 | 3.89 | 4.00 | 3.22 | 4.00 | 3.33 | 3.33 | 3.33 | 3.78 | 4.00 | 3.67 | 3.67 | 4.44 | 3.89 | 3.67 |

**TABLE 15.** Infrastructure feasibility – round 2 ratings.

| Participant | Text-to-Text Translation | Voice-to-Text Translation | Text-to-Voice Translation | Text or Voice to Image or Map | Offline Guide Maps | Multilingual AI ChatBot | PathFinder | LostPilgrim Tracker | MeetPoint Finder | Emergency Guide | Virtual Guide to Holy Sites | Scan and Understand | Charity Information & Locations | Transportation & Navigation | Haramain Explorer |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| P1 | 4 | 3 | 3 | 2 | 3 | 1 | 2 | 2 | 2 | 3 | 3 | 4 | 4 | 2 | 4 |
| P2 | 4 | 3 | 4 | 1 | 3 | 2 | 3 | 3 | 3 | 4 | 3 | 2 | 4 | 3 | 4 |
| P3 | 4 | 4 | 4 | 3 | 4 | 2 | 3 | 3 | 3 | 3 | 4 | 3 | 4 | 3 | 3 |
| P4 | 5 | 5 | 5 | 5 | 5 | 5 | 5 | 5 | 5 | 5 | 5 | 5 | 5 | 5 | 5 |
| P5 | 5 | 5 | 5 | 4 | 4 | 4 | 4 | 4 | 5 | 5 | 4 | 4 | 5 | 5 | 5 |
| P6 | 3 | 3 | 3 | 3 | 3 | 3 | 3 | 3 | 3 | 3 | 3 | 3 | 3 | 3 | 3 |
| P7 | 5 | 5 | 5 | 5 | 5 | 4 | 4 | 4 | 4 | 4 | 4 | 4 | 5 | 5 | 4 |
| P8 | 5 | 5 | 5 | 4 | 5 | 5 | 3 | 3 | 5 | 5 | 3 | 5 | 5 | 4 | 3 |
| P9 | 4 | 2 | 2 | 2 | 4 | 4 | 3 | 3 | 4 | 4 | 4 | 3 | 5 | 5 | 3 |
| **Mean** | 4.33 | 3.89 | 4.00 | 3.33 | 4.00 | 3.33 | 3.33 | 3.33 | 3.78 | 4.00 | 3.67 | 3.67 | 4.44 | 3.89 | 3.78 |

### *B. IMPLICATIONS*

The results of this study offer actionable insights for developers, policymakers, and technology planners involved in the digital transformation of the Hajj experience.

From a practitioner's perspective, features with high feasibility and priority scores should be prioritized for immediate implementation in mobile platforms addressing pilgrims challenges. For instance, integrating a feature-rich *Charity Info & Locations* module (score = 4.256) could streamline access to zakat and sadaqah services, while fostering trust through official verification mechanisms. Similarly, language-related tools such as *Text-to-Text Translation* and *Text-to-Voice Translation* offer both high utility and technical feasibility, making them ideal for initial rollouts. These tools directly address communication barriers and enhance inclusivity—especially for non-Arabic-speaking or elderly pilgrims. From a development standpoint, the prioritization results provide a roadmap for phased implementation. Features with moderate feasibility scores, such as *Haramain Explorer* or *MeetPoint Finder*, may be integrated in later stages after foundational feastures have been deployed and user feedback is collected. Developers can use this hierarchy to structure agile development cycles, minimizing risk while maximizing early impact. Policymakers and Hajj authorities

**TABLE 16.** Development feasibility – round 1 ratings.

| Participant | Text-to-Text Translation | Voice-to-Text Translation | Text-to-Voice Translation | Text or Voice to Image or Map | Offline Guide Maps | Multilingual AI ChatBot | PathFinder | LostPilgrim Tracker | MeetPoint Finder | Emergency Guidance | Virtual Guide to Holy Sites | Scan and Understand | Charity Information & Locations | Transportation & Navigation | Haramain Explorer |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| P1 | 4 | 3 | 3 | 2 | 3 | 2 | 2 | 2 | 2 | 2 | 3 | 3 | 3 | 2 | 3 |
| P2 | 5 | 3 | 4 | 2 | 3 | 3 | 3 | 3 | 3 | 3 | 3 | 2 | 4 | 4 | 3 |
| P3 | 5 | 5 | 5 | 4 | 4 | 2 | 3 | 3 | 3 | 3 | 4 | 3 | 4 | 4 | 3 |
| P4 | 5 | 5 | 5 | 5 | 5 | 5 | 5 | 5 | 5 | 5 | 5 | 5 | 5 | 5 | 5 |
| P5 | 5 | 5 | 5 | 3 | 3 | 4 | 5 | 5 | 5 | 5 | 5 | 5 | 5 | 5 | 4 |
| P6 | 2 | 3 | 3 | 5 | 5 | 4 | 4 | 2 | 2 | 2 | 2 | 2 | 2 | 2 | 2 |
| P7 | 5 | 4 | 4 | 3 | 4 | 3 | 3 | 3 | 3 | 3 | 3 | 4 | 4 | 3 | 3 |
| P8 | 5 | 5 | 5 | 4 | 4 | 5 | 4 | 4 | 5 | 5 | 3 | 5 | 5 | 2 | 2 |
| P9 | 5 | 5 | 5 | 5 | 4 | 5 | 3 | 2 | 4 | 5 | 5 | 3 | 3 | 5 | 5 |
| **Mean** | 4.56 | 4.22 | 4.33 | 3.67 | 3.89 | 3.67 | 3.56 | 3.22 | 3.56 | 3.67 | 3.67 | 3.56 | 3.89 | 3.56 | 3.33 |

**TABLE 17.** Development feasibility – round 2 ratings.

| Participant | Text-to-Text Translation | Voice-to-Text Translation | Text-to-Voice Translation | Text or Voice to Image or Map | Offline Guide Maps | Multilingual AI ChatBot | PathFinder | LostPilgrim Tracker | MeetPoint Finder | EmergencyGuide | Virtual Guide to Holy Sites | Scan and Understand | Charity Information and Locations | Transportation and Navigation | Haramain Explorer |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| P1 | 4 | 3 | 3 | 2 | 3 | 2 | 2 | 2 | 2 | 2 | 3 | 3 | 3 | 2 | 3 |
| P2 | 5 | 3 | 4 | 2 | 3 | 3 | 3 | 3 | 3 | 3 | 3 | 2 | 4 | 4 | 3 |
| P3 | 5 | 5 | 5 | 4 | 4 | 2 | 3 | 3 | 3 | 3 | 4 | 3 | 4 | 4 | 3 |
| P4 | 5 | 5 | 5 | 5 | 5 | 5 | 5 | 5 | 5 | 5 | 5 | 5 | 5 | 5 | 5 |
| P5 | 5 | 5 | 5 | 3 | 3 | 4 | 5 | 5 | 5 | 5 | 5 | 5 | 5 | 5 | 4 |
| P6 | 3 | 3 | 3 | 5 | 5 | 4 | 4 | 3 | 3 | 3 | 3 | 3 | 3 | 3 | 3 |
| P7 | 5 | 4 | 4 | 3 | 4 | 3 | 3 | 3 | 3 | 3 | 3 | 4 | 4 | 3 | 3 |
| P8 | 5 | 5 | 5 | 4 | 4 | 5 | 4 | 4 | 5 | 5 | 3 | 5 | 5 | 2 | 2 |
| P9 | 5 | 5 | 5 | 5 | 4 | 5 | 3 | 2 | 4 | 5 | 5 | 3 | 3 | 5 | 5 |
| **Mean** | 4.67 | 4.22 | 4.33 | 3.67 | 3.89 | 3.67 | 3.56 | 3.33 | 3.67 | 3.78 | 3.78 | 3.67 | 4.00 | 3.78 | 3.56 |

can leverage these insights to align digital infrastructure expansion with legal, logistical, and religious compliance frameworks. Priority features identified in this study serve as a rational basis for targeted budget allocation, regulation, and stakeholder engagement.

Beyond the Hajj setting, our study suggests generalizability to other mass gatherings where crowd density, intermittent connectivity, and context-specific rules dominate system behavior. In particular, offline-first, multilingual, and safety-oriented features—such as *Offline Guide Maps*, *Text-to-Text Translation*, *Text-to-Voice Translation*, *Transportation & Navigation*, *Emergency Guide*, and *MeetPoint Finder*—are likely to transfer with minimal adaptation, whereas advanced AR and data-intensive analytics—such as *AR PathFinder*, *Virtual Guide to Holy Sites*, *Lost Pilgrim Tracker*, and *Scan and Understand/Text/Voice to Image or Map Translation*—require careful tuning to local infrastructure and policy constraints. Many of the prioritized features (e.g., *Offline Guide Maps*, *Emergency Guide*, *Text-to-Text Translation*, and *MeetPoint Finder*) are directly applicable to Umrah, which shares the same locations but exhibits year-round demand rather than seasonal peaks. For non-Hajj mega-

**TABLE 18.** Regularity feasibility – round 1 ratings.

| Participant | Text-to-Text Translation | Voice-to-Text Translation | Text-to-Voice Translation | Text or Voice to Image or Map | Offline Guide Maps | Multilingual AI ChatBot | PathFinder | LostPilgrim Tracker | MeetPoint Finder | EmergencyGuide | Virtual Guide to Holy Sites | Scan and Understand | Charity Information and Locations | Transportation and Navigation | Haramain Explorer |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| P1 | 4 | 3 | 3 | 3 | 3 | 3 | 3 | 3 | 3 | 3 | 3 | 3 | 3 | 3 | 3 |
| P2 | 4 | 4 | 4 | 4 | 4 | 4 | 4 | 4 | 4 | 4 | 4 | 4 | 4 | 4 | 4 |
| P3 | 1 | 1 | 1 | 1 | 1 | 1 | 2 | 2 | 2 | 1 | 2 | 2 | 1 | 1 | 2 |
| P4 | 5 | 5 | 5 | 5 | 5 | 5 | 5 | 5 | 5 | 5 | 5 | 5 | 5 | 5 | 5 |
| P5 | 5 | 5 | 5 | 4 | 5 | 5 | 5 | 5 | 5 | 5 | 4 | 5 | 5 | 5 | 4 |
| P6 | 3 | 2 | 2 | 3 | 3 | 3 | 3 | 3 | 3 | 3 | 3 | 3 | 3 | 3 | 3 |
| P7 | 5 | 4 | 5 | 4 | 5 | 3 | 3 | 3 | 3 | 3 | 3 | 5 | 3 | 3 | 3 |
| P8 | 5 | 5 | 5 | 4 | 5 | 5 | 4 | 4 | 4 | 5 | 5 | 5 | 5 | 4 | 4 |
| P9 | 4 | 3 | 5 | 2 | 5 | 5 | 4 | 4 | 4 | 4 | 4 | 3 | 5 | 5 | 5 |
| **Mean** | 3.89 | 3.56 | 3.89 | 3.33 | 4.00 | 3.78 | 3.67 | 3.67 | 3.67 | 3.67 | 3.67 | 3.89 | 3.78 | 3.67 | 3.67 |

**TABLE 19.** Regularity Feasibility – round 2 ratings.

| Participant | Text-to-Text Translation | Voice-to-Text Translation | Text-to-Voice Translation | Text or Voice to Image or Map | Offline Guide Maps | Multilingual AI ChatBot | PathFinder | LostPilgrim Tracker | MeetPoint Finder | EmergencyGuide | Virtual Guide to Holy Sites | Scan and Understand | Charity Information and Locations | Transportation and Navigation | Haramain Explorer |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| P1 | 3 | 3 | 3 | 3 | 3 | 3 | 3 | 3 | 3 | 3 | 3 | 3 | 3 | 3 | 3 |
| P2 | 4 | 4 | 4 | 4 | 4 | 4 | 4 | 4 | 4 | 4 | 4 | 4 | 4 | 4 | 4 |
| P3 | 5 | 5 | 5 | 5 | 5 | 5 | 4 | 4 | 4 | 5 | 4 | 4 | 5 | 5 | 4 |
| P4 | 5 | 5 | 5 | 5 | 5 | 5 | 5 | 5 | 5 | 5 | 5 | 5 | 5 | 5 | 5 |
| P5 | 5 | 5 | 5 | 4 | 5 | 5 | 5 | 5 | 5 | 5 | 4 | 5 | 5 | 5 | 4 |
| P6 | 3 | 3 | 3 | 3 | 3 | 3 | 3 | 3 | 3 | 3 | 3 | 3 | 3 | 3 | 3 |
| P7 | 5 | 4 | 5 | 4 | 5 | 3 | 3 | 3 | 3 | 3 | 3 | 5 | 3 | 3 | 3 |
| P8 | 5 | 5 | 5 | 4 | 5 | 5 | 5 | 4 | 4 | 5 | 5 | 5 | 5 | 5 | 4 |
| P9 | 4 | 3 | 5 | 2 | 5 | 5 | 4 | 4 | 4 | 4 | 4 | 3 | 5 | 5 | 5 |
| **Mean** | 4.33 | 4.11 | 4.44 | 3.78 | 4.44 | 4.22 | 4.00 | 3.89 | 3.89 | 4.11 | 3.89 | 4.11 | 4.22 | 4.22 | 3.89 |

events (e.g., other pilgrimages, festivals, sports tournaments), transfer mainly requires re-parameterizing geofences, substituting local transport/ticketing APIs, and updating language packs and emergency protocols. For effective cross-cultural adoption, interfaces should support both RTL and LTR scripts, use clear icon-forward navigation for low-literacy users, provide multimodal guidance (text, audio, visual cues), and implement transparent consent with short retention for sensitive data (e.g., precise location). These practices guide localization for diverse pilgrim groups and other international audiences. These findings also align with recent work showing how smart tourism technologies can foster sustainable development and visitor loyalty in religious tourism settings [66].

### C. LIMITATIONS AND FUTURE WORK

While this study presents a systematic and expert-validated methodology for identifying and evaluating mobile application features tailored to the Hajj context, several limitations must be acknowledged.

**TABLE 20.** Environmental feasibility – round 1 ratings.

| Participant | Text-to-Text Translation | Voice-to-Text Translation | Text-to-Voice Translation | Text or Voice to Image or Map | Offline Guide Maps | Multilingual AI ChatBot | PathFinder | LostPilgrim Tracker | MeetPoint Finder | EmergencyGuide | Virtual Guide to Holy Sites | Scan and Understand | Charity Information and Locations | Transportation and Navigation | Haramain Explorer |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| P1 | 3 | 3 | 3 | 3 | 3 | 3 | 3 | 3 | 3 | 3 | 3 | 3 | 3 | 3 | 3 |
| P2 | 2 | 3 | 4 | 4 | 4 | 3 | 4 | 4 | 4 | 4 | 2 | 3 | 3 | 2 | 3 |
| P3 | 4 | 2 | 3 | 2 | 5 | 3 | 5 | 5 | 3 | 4 | 3 | 3 | 4 | 4 | 4 |
| P4 | 5 | 5 | 5 | 5 | 5 | 5 | 5 | 5 | 5 | 5 | 5 | 5 | 5 | 5 | 5 |
| P5 | 5 | 5 | 5 | 3 | 5 | 5 | 4 | 4 | 4 | 5 | 5 | 4 | 5 | 3 | 5 |
| P6 | 2 | 2 | 2 | 2 | 2 | 2 | 2 | 3 | 3 | 3 | 2 | 2 | 4 | 4 | 4 |
| P7 | 5 | 5 | 3 | 4 | 4 | 5 | 3 | 3 | 3 | 3 | 5 | 3 | 5 | 5 | 5 |
| P8 | 1 | 2 | 1 | 4 | 3 | 1 | 4 | 5 | 4 | 5 | 3 | 3 | 3 | 4 | 5 |
| P9 | 2 | 4 | 4 | 2 | 2 | 3 | 3 | 2 | 4 | 4 | 4 | 3 | 3 | 2 | 5 |
| **Mean** | 3.22 | 3.44 | 3.33 | 3.22 | 3.67 | 3.33 | 3.67 | 3.78 | 3.67 | 4.00 | 3.56 | 3.22 | 3.89 | 3.56 | 4.33 |

**TABLE 21.** Environmental feasibility – round 2 ratings.

| Participant | Text-to-Text Translation | Voice-to-Text Translation | Text-to-Voice Translation | Text or Voice to Image or Map | Offline Guide Maps | Multilingual AI ChatBot | PathFinder | LostPilgrim Tracker | MeetPoint Finder | EmergencyGuide | Virtual Guide to Holy Sites | Scan and Understand | Charity Information and Locations | Transportation and Navigation | Haramain Explorer |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| P1 | 3 | 3 | 3 | 3 | 3 | 3 | 3 | 3 | 3 | 3 | 3 | 3 | 3 | 3 | 3 |
| P2 | 2 | 3 | 4 | 4 | 4 | 3 | 4 | 4 | 4 | 4 | 2 | 3 | 3 | 2 | 3 |
| P3 | 4 | 2 | 3 | 2 | 5 | 3 | 5 | 5 | 3 | 4 | 3 | 3 | 4 | 4 | 4 |
| P4 | 5 | 5 | 5 | 5 | 5 | 5 | 5 | 5 | 5 | 5 | 5 | 5 | 5 | 5 | 5 |
| P5 | 5 | 5 | 5 | 3 | 5 | 5 | 4 | 4 | 4 | 5 | 5 | 4 | 5 | 3 | 5 |
| P6 | 3 | 3 | 3 | 3 | 3 | 3 | 3 | 3 | 3 | 3 | 3 | 3 | 4 | 4 | 4 |
| P7 | 5 | 5 | 3 | 4 | 4 | 5 | 3 | 3 | 3 | 3 | 5 | 3 | 5 | 5 | 5 |
| P8 | 1 | 4 | 2 | 4 | 4 | 3 | 5 | 5 | 5 | 5 | 3 | 3 | 5 | 4 | 5 |
| P9 | 2 | 4 | 4 | 2 | 2 | 3 | 3 | 2 | 4 | 4 | 4 | 3 | 3 | 2 | 5 |
| **Mean** | 3.33 | 3.78 | 3.56 | 3.33 | 3.89 | 3.67 | 3.89 | 3.78 | 3.78 | 4.00 | 3.67 | 3.33 | 4.11 | 3.56 | 4.33 |

First, the evaluation relied on a relatively small panel of nine domain experts. Although the Delphi method enhances the validity of consensus-based assessments and helps mitigate individual bias, future work should consider engaging a larger expert group to improve the generalizability and representativeness of the results. Second, the study adopted an expert-driven evaluation approach without direct involvement of end users (i.e., pilgrims). At this stage, the features remain at a conceptual/design-validation level rather than empirically tested with end users. This approach is common in the literature, where researchers routinely build expert consensus on features, content, or assessment frameworks before undertaking end-user testing [28], [67], [68], [69]. As such, while the identified features are considered feasible and contextually appropriate by experts, their actual usability, relevance, and acceptability among pilgrims remain untested. Future research should incorporate participatory design methods, usability evaluations, and field-based deployment studies to capture end-user perspectives and iteratively refine the feature set.

Third, the current evaluation focused primarily on feasibility across infrastructure, development, regulatory, and environmental dimensions. We prioritized these dimensions because they determine to how much extent a feature can be buildable, approvable, and functional under real Hajj conditions. However, additional dimensions could further strengthen the evaluation of features—specifically, *cost-effectiveness*, *scalability to peak Hajj volumes*, *cybersecurity*, and *data privacy*. These were outside the scope of the current work but are planned as future extensions. Lastly, empirical validation with real pilgrims—including pilot implementation, usability testing, and performance evaluation in real Hajj settings—would further strengthen the work. However, due to logistical, regulatory, and ethical constraints associated with field trials during Hajj, such testing was beyond the scope of the present study. To address this, we will (i) develop a functional prototype that implements the prioritized feature set, (ii) conduct usability testing with prospective pilgrims in controlled settings (e.g., Hajj training centers and orientation sessions), and (iii) run pilot studies during Umrah as a practical proxy for the Hajj environment before scaling to full Hajj deployment in the future. These enhancements will help strengthen both the practical impact and theoretical underpinnings of smart pilgrimage technologies.

## VII. CONCLUSION

This study employed a Delphi-guided methodology to identify, assess and prioritize mobile application features supporting Hajj pilgrims. The process began by mapping key pilgrim challenges and proposing 15 features. A panel of nine domain experts then evaluated these features over two Delphi rounds to assess their relevance in addressing pilgrims challenges. After reaching consensus on feature inclusion, a second phase rated each feature across four feasibility dimensions—infrastructure, development, regulatory, and environmental—to guide prioritization for implementation.

The study addresses a critical gap in the literature by systematically linking identified pilgrim challenges to proposed technological features, validating these through expert consensus, and offering a prioritization framework to guide implementation. This approach not only yields a validated feature set but also provides clear guidance for developers, policymakers, and technology planners working to improve the Hajj experience through mobile platforms. As the Hajj continues to grow in scale and complexity, adopting such proposed features will be essential to ensure that digital interventions are not only innovative, but also impactful. Moreover, the proposed features are scalable to Umrah and other mass gatherings, while successful cross-cultural adoption will depend on careful localization, regulatory compliance, and infrastructure adaptation to support diverse international audiences.

## DECLARATIONS

### A. AI USE IN THE WRITING PROCESS

The authors declare that Generative-AI tools were used for grammar checking, and rephrasing. Additionally, they confirm that ChatGPT was employed to generate conceptual images of proposed features through image prompts. They affirm that the core ideas and contributions presented in this work are novel.

### B. COMPETING INTERESTS

The authors declare that they have no financial or other conflicts of interest with any external entities or organizations that could inappropriately influence or bias the findings of this work.

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